Unnamed_Unit

MEET YOUR COMPUTER: EXPLORING HARDWARE, SOFTWARE, AND SYSTEMS

How Robotics Works?

Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality. Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality**Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality. Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality*****Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality. Starting your journey to a successful career in Germany requires a clear path and a guiding hand. At ABC Global Employment, we’ve crafted a straightforward, supportive process to make every step as smooth as possible. From the initial consultation to settling into your new role, our team is here to provide personalized guidance and expertise. We handle the complexities so you can focus on the exciting opportunities ahead. Together, we’ll turn your dreams of working in Germany into a reality*****

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Gallery Images

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PDF Documents

PDF Document 1: 1731654515711-STEAM Journey - Grade 4 (1st Edition).pdf

This PDF document is included in the unit materials.

PDF Document 2: 1731749917349-Unit 1 Cracking the Binary Code.pdf

This PDF document is included in the unit materials.

YouTube Links

Grade 4

SCIENCE
TECHNOLOGY
ENGINEERING
ARTS
MATHEMATICS
ROBOTICS
CODING
STEAM EDUCATION
AUTOMATION

STEAM JOURNEY WITH MODI+

TERM 1

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STEAM JOURNEY

Grade 4

Term 1

A Message from the Global Educational Team Nanoskool

To educators across the globe,

We want to express our heartfelt appreciation to each one of you who utilize the STEAM Journey series to inspire, educate, and empower students. At Nanoskool, we are passionate about advancing STEAM education because we believe it unlocks countless possibilities and paves the way for students to explore new frontiers. Your unwavering commitment to teaching transforms our mission into a tangible reality, nurturing the next generation of innovators and leaders. We are grateful for your dedication, your belief in the potential of every student, and your role in sparking curiosity and fostering a love of learning that extends far beyond the classroom walls.

Thank you for all that you do.

Warm regards,
Nanoskool

GRADE 4

STEAM JOURNEY

TERM 1

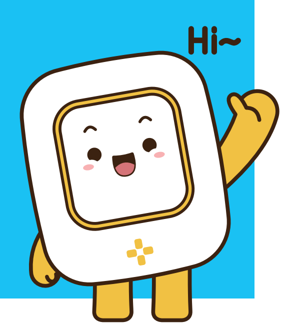
A Message from the Global Educational Team at Luxrobo Co., Ltd.

To educators across the globe,

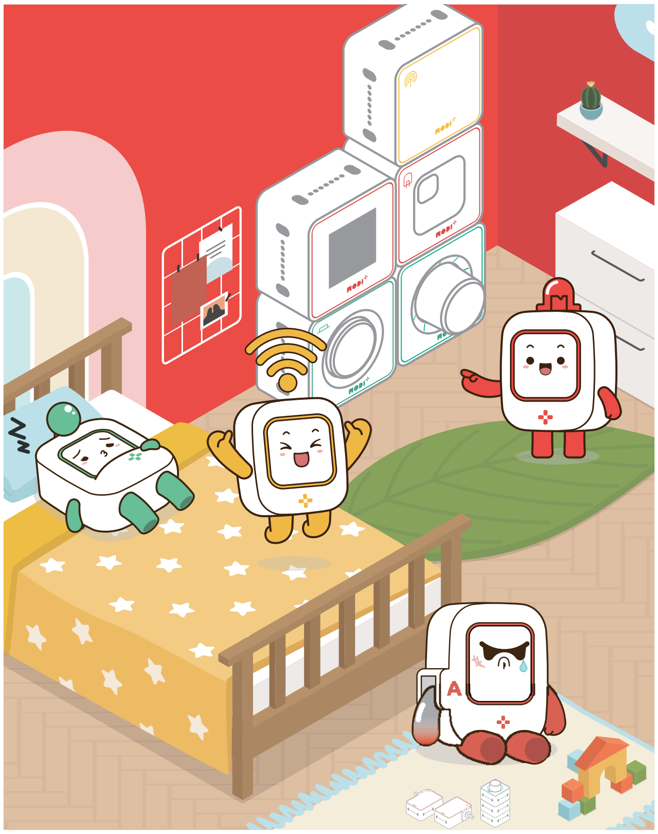
We extend our deepest gratitude to all of you who guide, inspire, and teach using the STEAM Journey series. At Luxrobo, we are committed to advancing STEAM education because we believe in its power to broaden horizons and open new opportunities for students everywhere. Your dedication to teaching transforms this commitment into reality, paving the way for future innovators and leaders. Thank you for your passion and for believing in the potential of every student. Together, we can ignite curiosity and foster learning that reaches beyond the classroom.

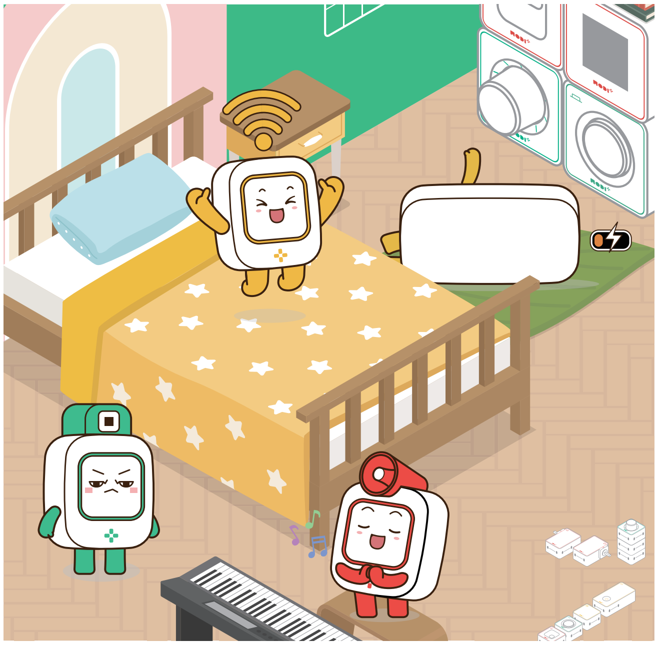
With sincere thanks,

Luxrobo Global Educational Team



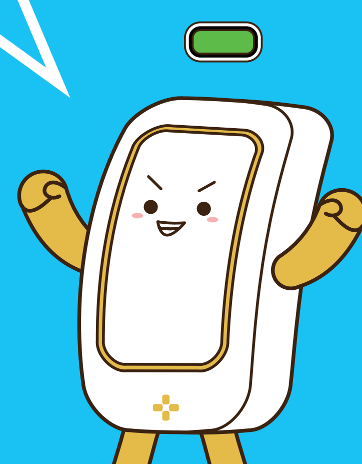
BRIEF CONTENTS





GRADE 4

STEAM JOURNEY



CHAPTER 1

COMPUTER SCIENCE & TECHNOLOGY

TOPIC MEET YOUR COMPUTER: EXPLORING HARDWARE, SOFTWARE, AND SYSTEMS

CONTENT

UNIT 1 LEARNING ABOUT BASIC HARDWARE

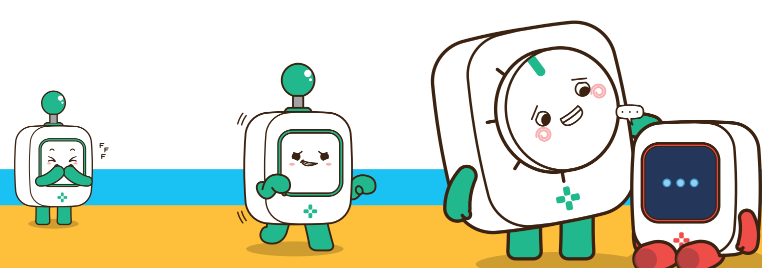
Description: Explore the important parts of a computer, like the monitor, keyboard, mouse, and CPU. Discover what each part does and how they work together to make your computer run.

UNIT 2 DISCOVERING THE OPERATING SYSTEM

Description: Learn about the operating system, the software that helps you control your computer. See how it manages everything from opening programs to running games.

UNIT 3 EXPLORING SOFTWARE

Description: Find out how software makes your computer useful and fun! Learn about different types of software and how they help you create, play, and learn.



CHAPTER 1

MEET YOUR COMPUTER: EXPLORING HARDWARE, SOFTWARE, AND SYSTEMS

What You Will Learn

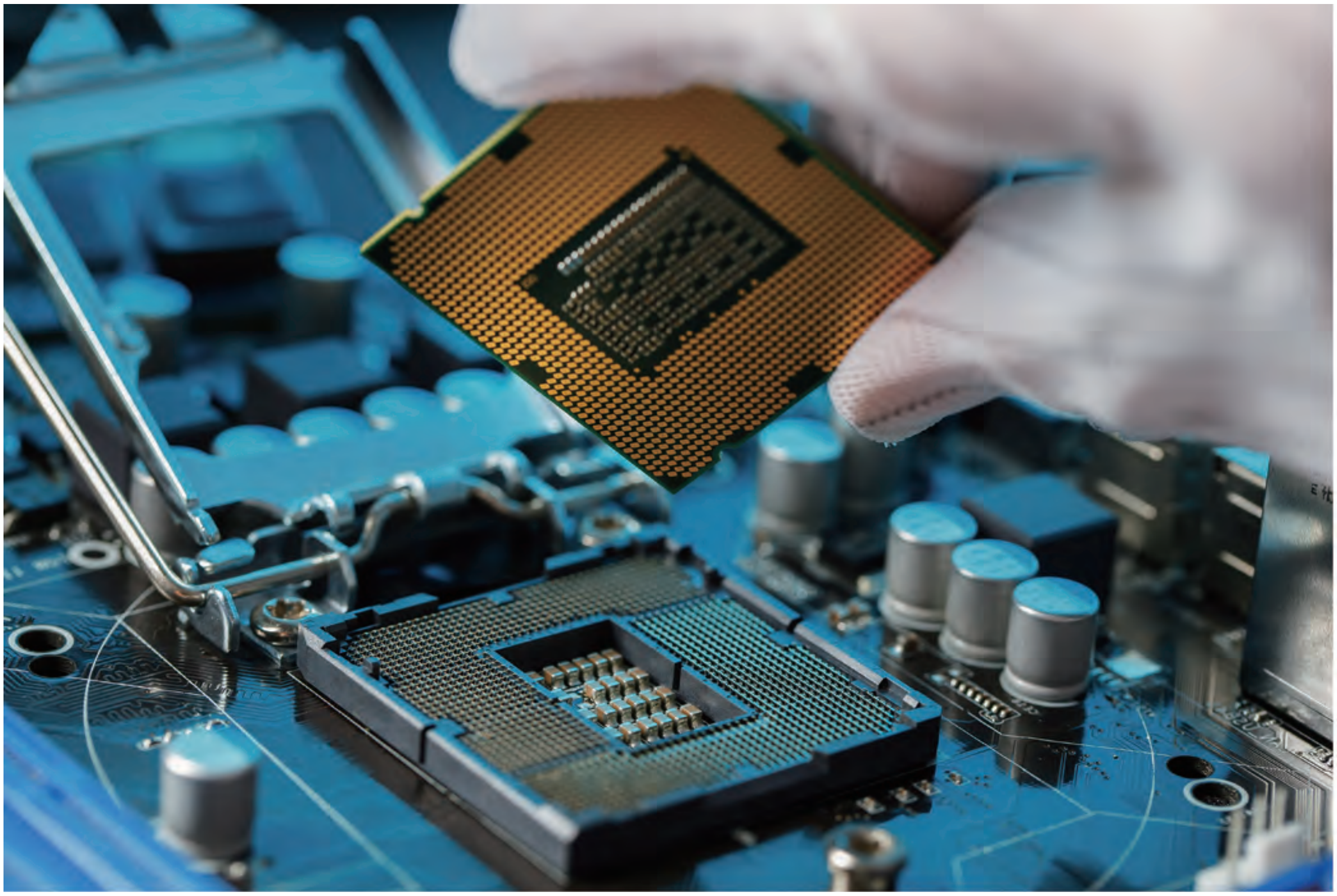
- What the main parts of a computer are and what they do.
- How the operating system helps you use your computer.
- How different software programs can help you create, play, and learn.

Learning Objectives

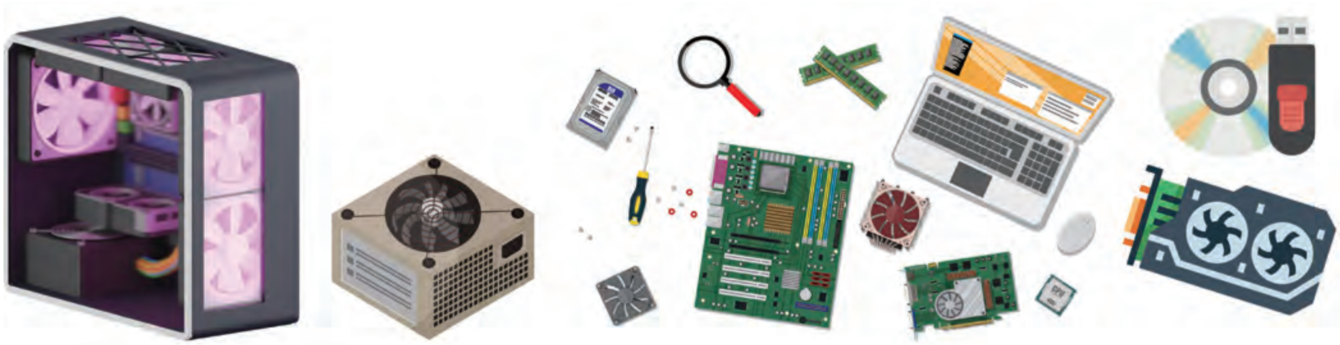
- Identify the key hardware components of a computer and describe their functions.
- Understand how the operating system allows you to interact with your computer.
- Explore and use different types of software to perform tasks and have fun.

UNIT 1 LEARNING ABOUT BASIC HARDWARE

Basic Hardware



Welcome to Unit 1! In this unit, you will start your journey into the world of computers by learning about the basic hardware that makes a computer work. Hardware refers to the physical parts of the computer that you can see and touch. By the end of this unit, you'll know the names of these parts and understand how they work together to make your computer run smoothly.



UNDERSTANDING ANGLES

MONITOR

This is the screen that shows you everything you're doing on the computer. It displays pictures, videos, and text.

KEYBOARD

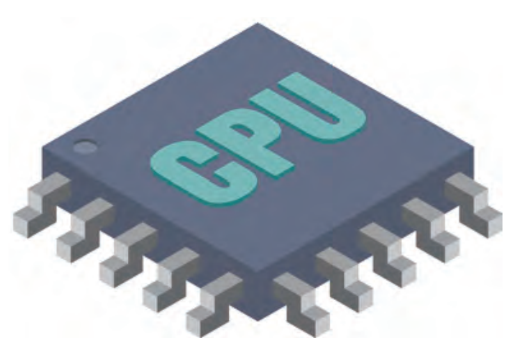
You use this to type letters, numbers, and symbols. It helps you talk to the computer.

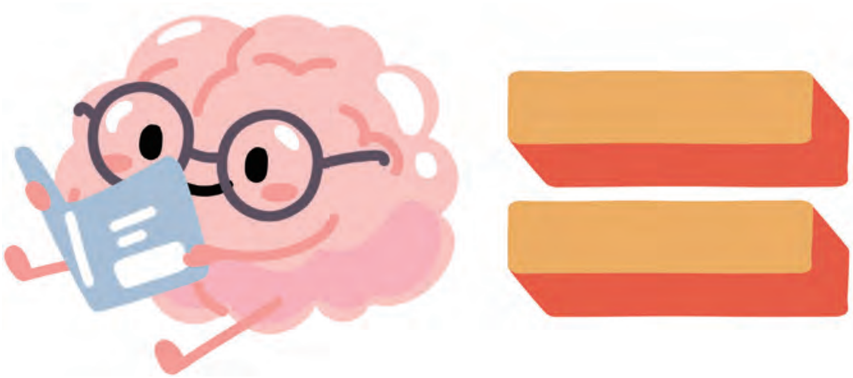
MOUSE

This small device helps you move around on the screen and click on things.

CPU (CENTRAL PROCESSING UNIT)

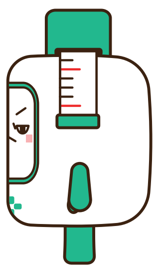
The CPU is like the brain of the computer. It controls everything and makes sure all the parts work together.





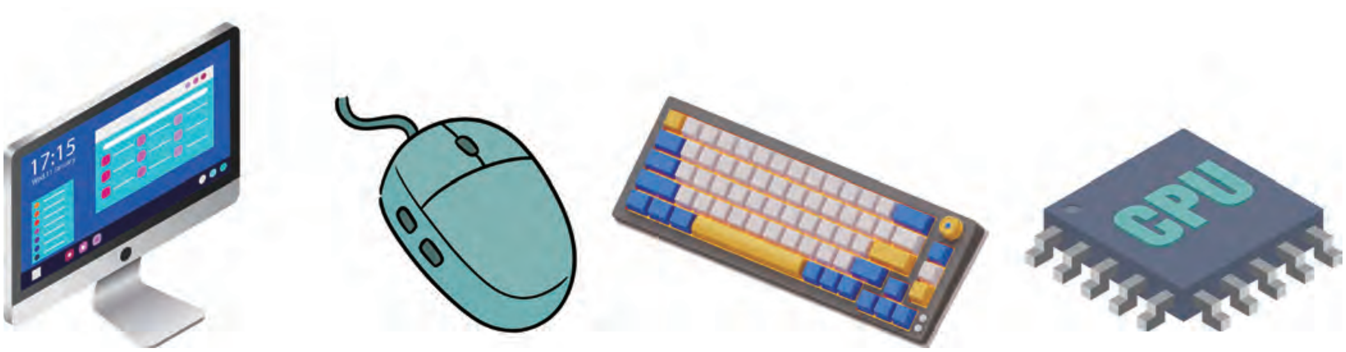
DID YOU KNOW?

Without the CPU, the computer wouldn't know what to do! That's why it's often called the brain of the computer.



Remember the Components We Learned!

Below is a picture of a desktop setup. Remember the components we learned about on the previous page! Share what each component is with your classmates.



Activity 1

Let's Label the Parts of a Computer!

Now it's your turn! Use the words you've learned (monitor, keyboard, mouse, CPU) to label the parts of the computer in the picture.

1 • Look at the picture of the computer.

2 • Write the correct name next to each part.

3 • Color the parts to make your diagram unique!

Name : ______________

Name : ______________

Name : ______________

Name : ______________

Group Discussion

Discussion question 1

What do you see on the monitor? Why do you need it?

Discussion question 2

How does the keyboard help you? What can you do with it?

Discussion question 3

What do you see on the monitor? Why do you need it?

Discussion question 4

How does the keyboard help you? What can you do with it?

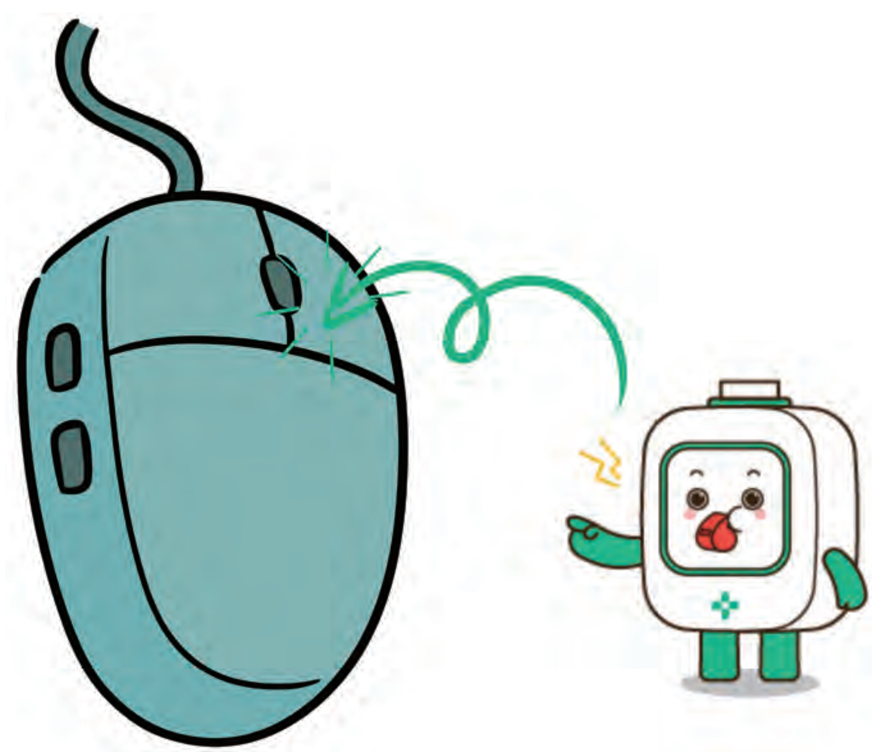
Activity 2

Let's Label the Parts of a Computer!

Now that you've learned about the different parts of a computer, it's time to use them in real life! You'll be in a computer lab, where each of you (or your group) will have a desktop or laptop to explore. Let's see how these parts work when you use them.

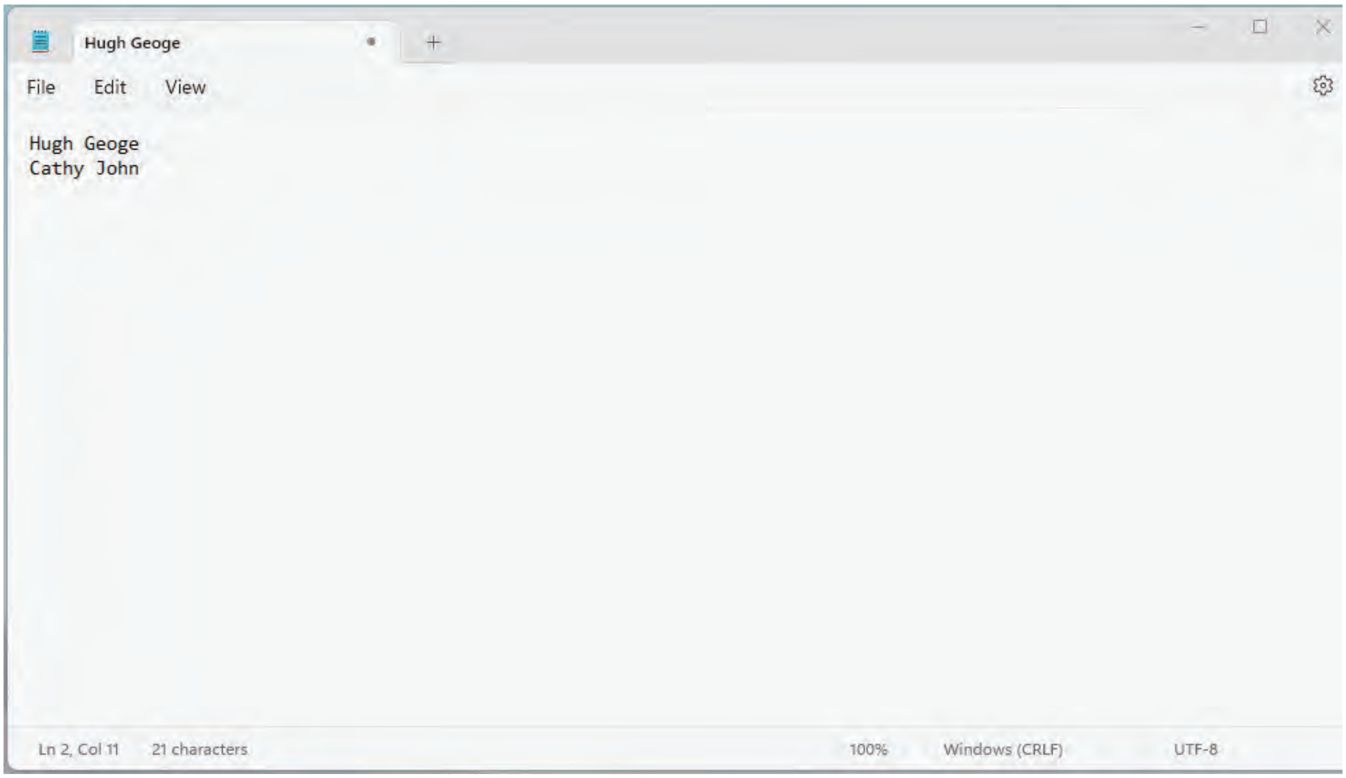
1. CLICK THE RIGHT BUTTON ON YOUR MOUSE

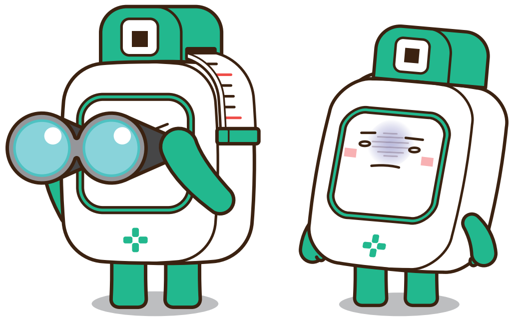
- What to Do: Find the right button on your mouse and click it.
- What Happens: Watch what appears on the screen. Does a menu pop up? Talk about what options you see.



2. FIND THE MEMO APP AND TYPE YOUR NAME

- What to Do: Open the memo or notepad app on your computer.
- Using the Keyboard: Type your name in the app. Watch the letters appear on the screen as you type!
- Try This: Erase your name using the backspace key, and then type it again.





3. USE THE KEYBOARD SHORTCUTS

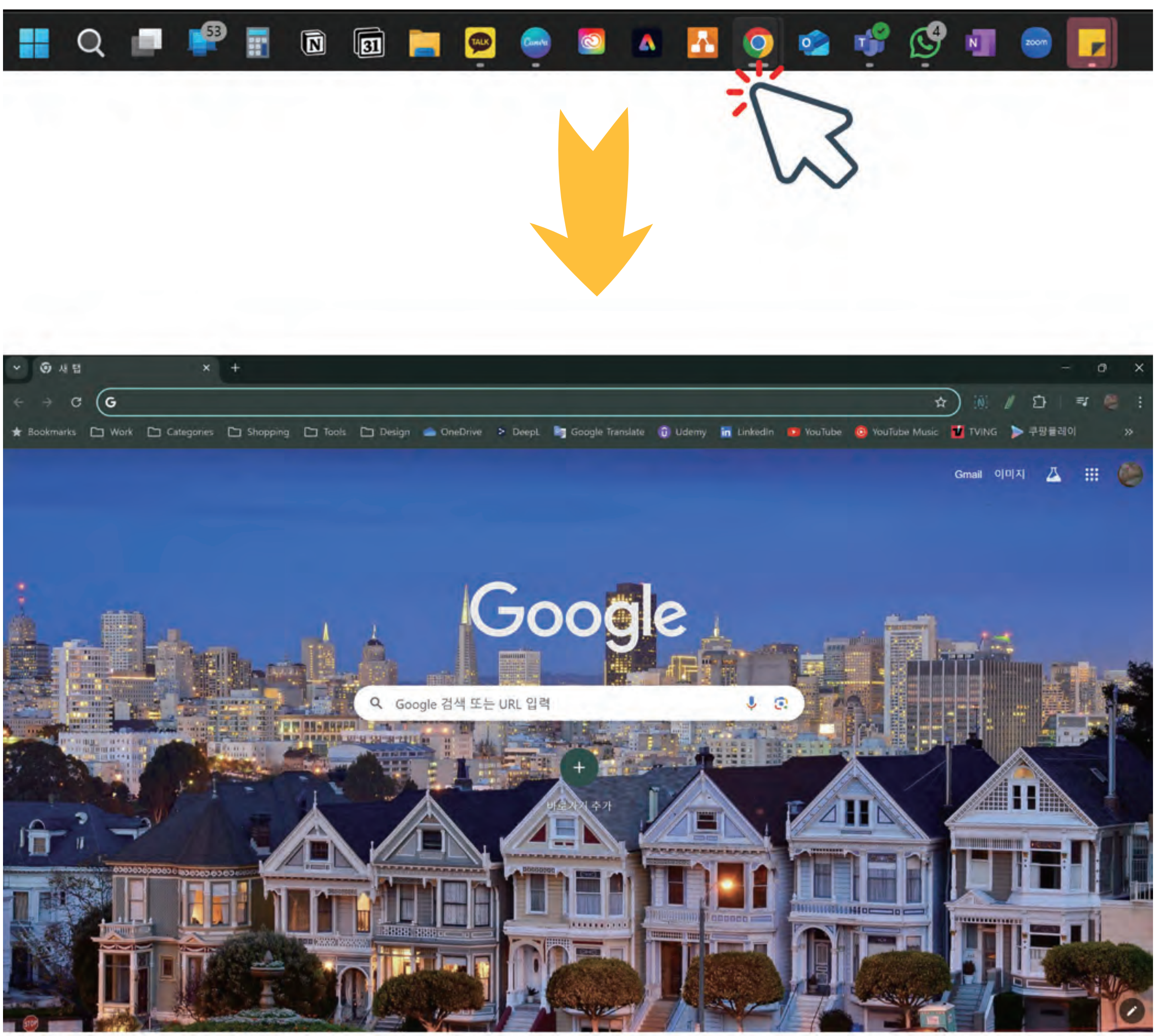
- What to Do: Try pressing "Ctrl + C" to copy something and "Ctrl + V" to paste it.
- Experiment: Type some text, copy it, and then paste it somewhere else in the memo app.

4. TURN OFF THE MONITOR, THEN TURN IT ON AGAIN

- What to Do: Find the power button on your monitor and press it to turn the screen off.
- Next Step: Wait a few seconds, then press the power button again to turn it back on.
- What Happens: Notice how the screen goes dark and then comes back to life.

5. OPEN A PROGRAM USING THE MOUSE

- What to Do: Use the mouse to double-click on an icon to open a program (like a web browser or a game).
- Explore: Once the program is open, explore what you can do with it. Can you move the window around the screen? Can you close it?



Quick Quiz

Answer these questions to check what you've learned about computer parts. Circle the correct answer.

Quiz Questions:

1 What part of the computer shows you what's happening on the screen?

Ⓐ Mouse
Ⓑ Monitor
Ⓒ CPU

2 What do you use to type letters and numbers on the computer?

Ⓐ Monitor
Ⓑ CPU
Ⓒ Keyboard

3 Which part of the computer is known as the brain?

Ⓐ Mouse
Ⓑ CPU
Ⓒ Monitor

4 What do you use to type letters and numbers on the computer?

Ⓐ Move around and click on things
Ⓑ Type words
Ⓒ Show pictures

CHAPTER 2

INTRODUCTION TO THE MODI MASTER KIT

TOPIC **WELCOME TO THE WORLD OF MODI!**

CONTENT

UNIT 1 GETTING TO KNOW THE MODI MASTER KIT

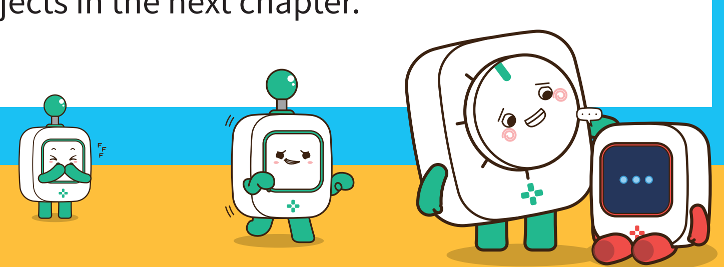
Description: Learn about the MODI Master Kit and explore the different modules inside. Discover what each module does and how it can be used in projects.

UNIT 2 FIRST STEPS WITH MODI: UNPLUGGED ACTIVITIES

Description: Dive deeper into the MODI modules. Learn how sensor modules, actuator modules, and network modules work, and how they can be connected to create exciting projects.

UNIT 3 NAVIGATING THE CODE EDITOR

Description: Learn how to use the MODI Code Editor to start programming your modules. Discover how to connect and control your modules through coding, and get ready to build your first projects in the next chapter.



CHAPTER 2

INTRODUCTION TO THE MODI MASTER KIT

What You Will Learn

- **MODI Kit Basics:** Learn what the MODI Master Kit is and how it can be used to create cool projects.
- **Module Functions:** Explore the different types of modules in the kit, like sensors, actuators, and network modules, and see how they work.
- **Connecting Modules:** Discover how to connect modules together to build simple systems.
- **Preparing for Projects:** Prepare yourself for programming and building projects with the MODI Kit in the next chapter.

Learning Objectives

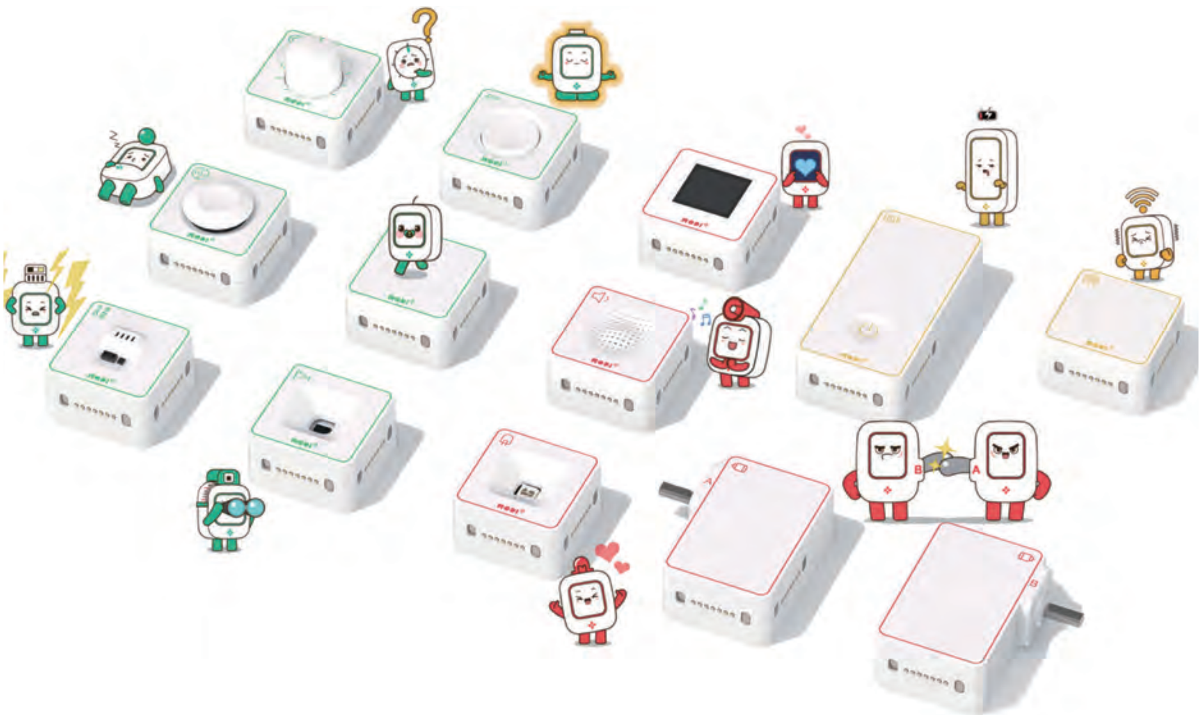
- **Understand the MODI Kit:** Learn what the MODI Master Kit is and why it's a great tool for learning about technology and programming.
- **Identify and Explain Modules:** Identify the different modules in the MODI Kit and explain what each one does.
- **Connect and Create:** Find out how to connect different modules to start building simple systems.
- **Prepare for Programming:** Get ready to start programming with MODI by understanding how the modules work together and how to take care of your kit.

UNIT 1 GETTING TO KNOW THE MODI MASTER KIT

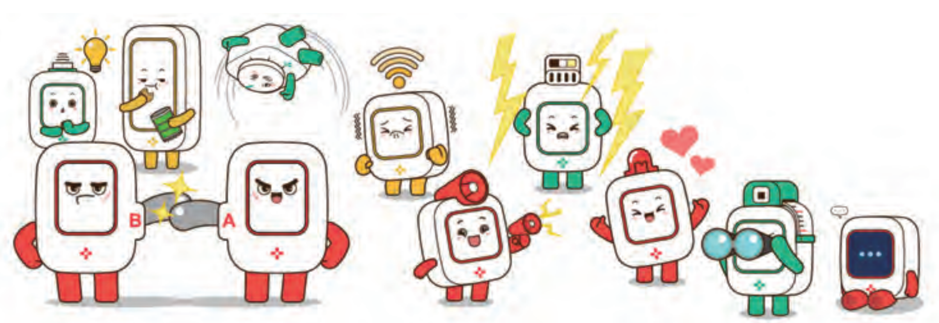
What Is the MODI Master Kit?

WELCOME TO THE MODI MASTER KIT!

This kit is like a set of magical building blocks that can help you create all sorts of amazing projects.



Each block, called a module, has a special job. Some modules can sense things like light or sound, while others can move objects or show lights. Before you start building, it's important to get to know each of these modules and learn how they work.

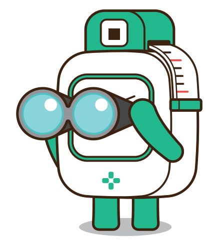


Discovering the MODI Master Kit

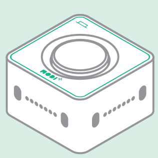
Welcome to the world of MODI! Inside your MODI Master Kit, you'll find colorful modules. Each one has a special job, and they're fun and easy to use. These modules are the building blocks for creating gadgets and robots.

Understanding MODI Modules by Color (3 Types of Modules)

These modules are like senses for your device. They gather information about sounds, temperature, distance, and more, just like our senses help us understand the world.

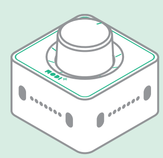


INPUT MODULES



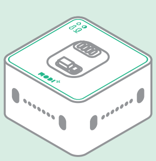
Button

Senses the status of the button



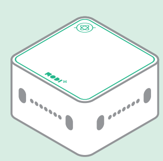
Dial

Senses the rotation of the dial



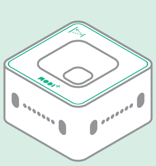
Environment

Senses temperature, humidity, illuminance, and sound



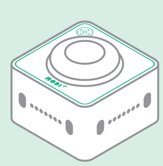
IMU

Senses slopes (Angles)



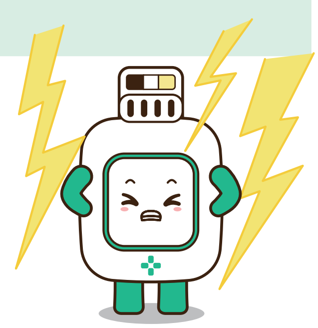
ToF

Senses distance

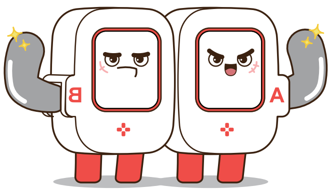


Joystick

Senses the direction of the stick

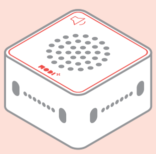


RED: THE COLOR OF ACTION WITH OUTPUT MODULES

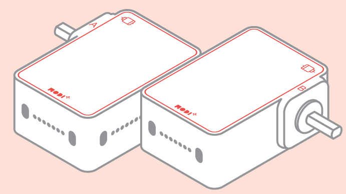


Red MODI modules are the doers. They take information and make things happen, like lighting up, playing sounds, or moving parts.

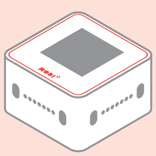
OUTPUT MODULES


Speaker

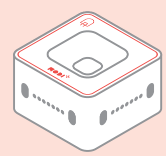
Make sounds


Motor A, B

Rotates


display

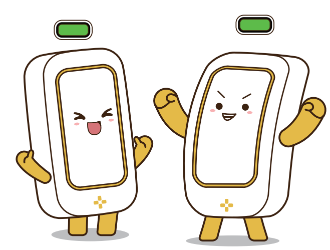
Shows letters and images


LED

Emits light

YELLOW: THE FOUNDATION WITH SETUP MODULES

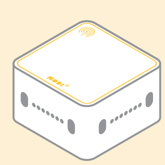
Yellow MODI modules are super important. They give power and connections so all the other modules can work together smoothly.



SETUP MODULES


Battery

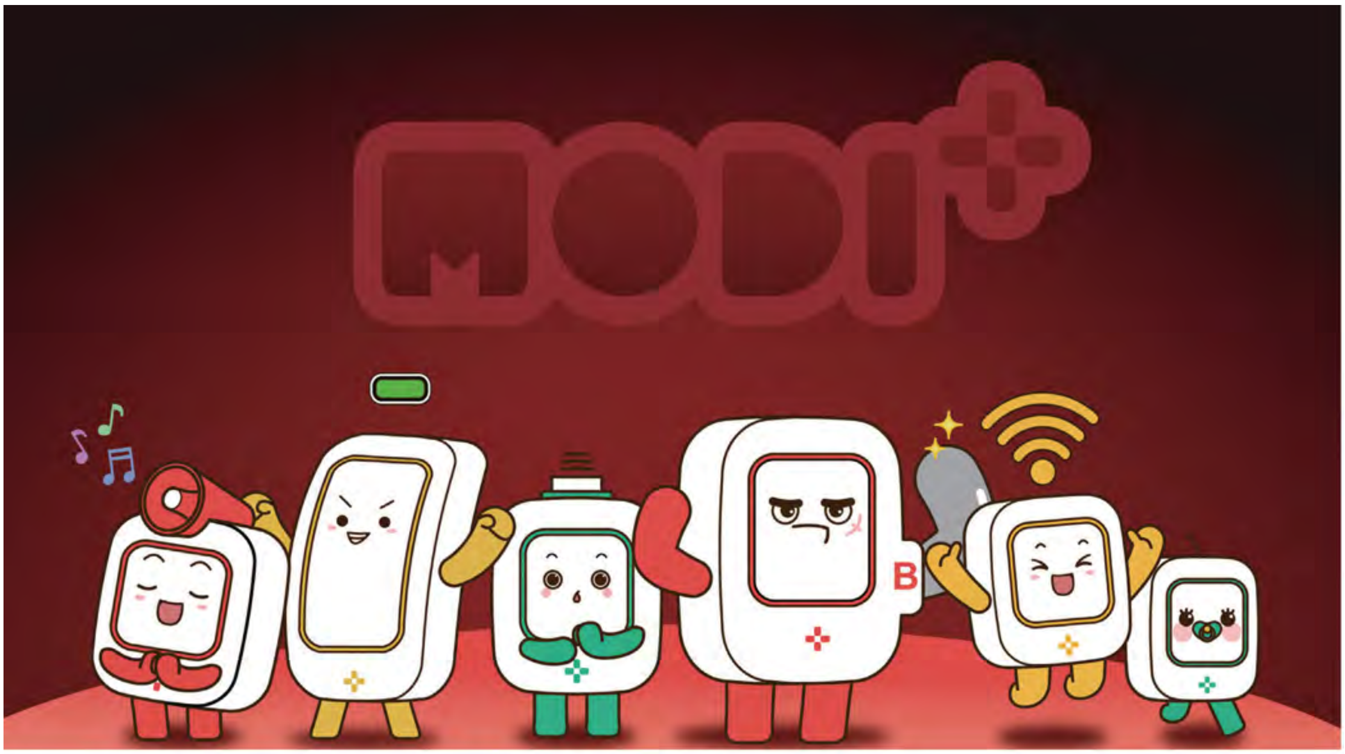
Supplies energy to the modules


Network

Connects modoules to other modules, PCs, tablets, and phones

Bringing It All Together

MODI modules work together just like parts of a computer. Each module has a special job, and when they work together, amazing things happen. No single part can do everything on its own.



Think about your smartphone. It has many parts, like a screen that shows pictures and sensors that detect light. Each part does its job, and together, they make your phone work. The same goes for MODI projects—it's all about teamwork!

UNIT 2 FIRST STEPS WITH MODI: UNPLUGGED ACTIVITIES

In robotics, an "unplugged activity" is like a puzzle or game you do without a computer. Before we start using our MODI modules, we'll do some unplugged activities to warm up our brains. This will help us think like programmers and understand how our MODI robots will work.

Unplugged

Activity1 Environment Meter

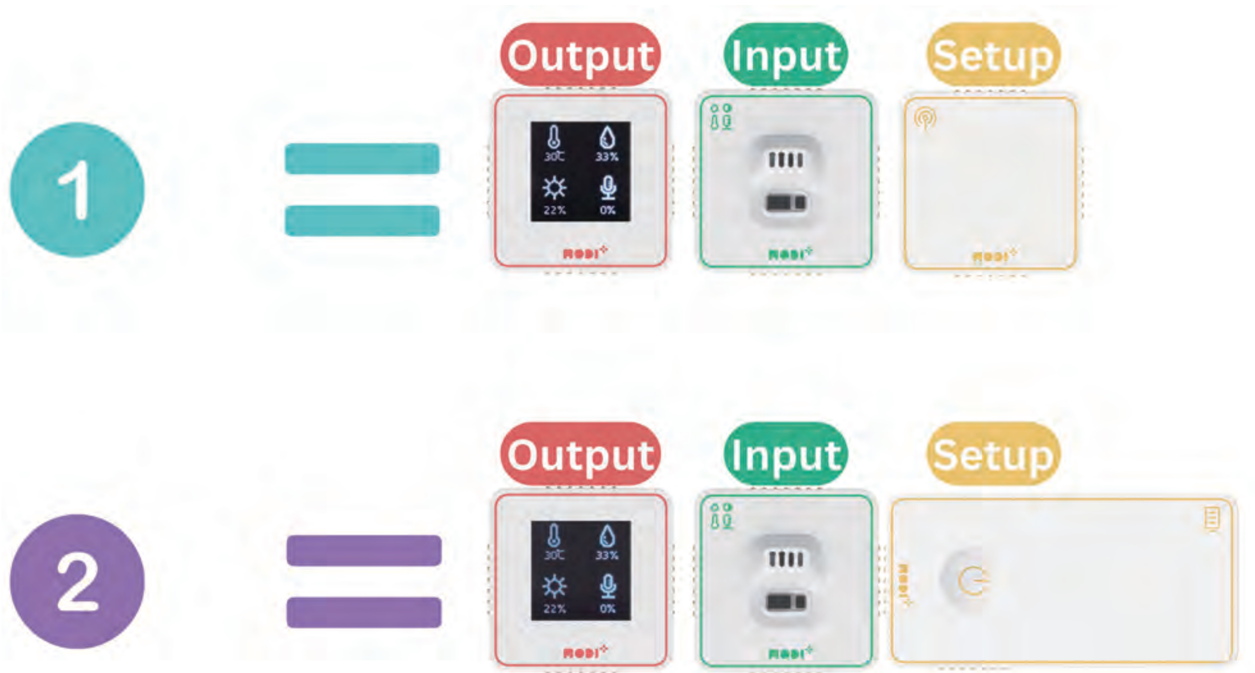
Instructions:

1. CONNECTING YOUR MODI MODULES Just bring the modules close together, and they'll snap into place like magic, thanks to the magnets on their sides.

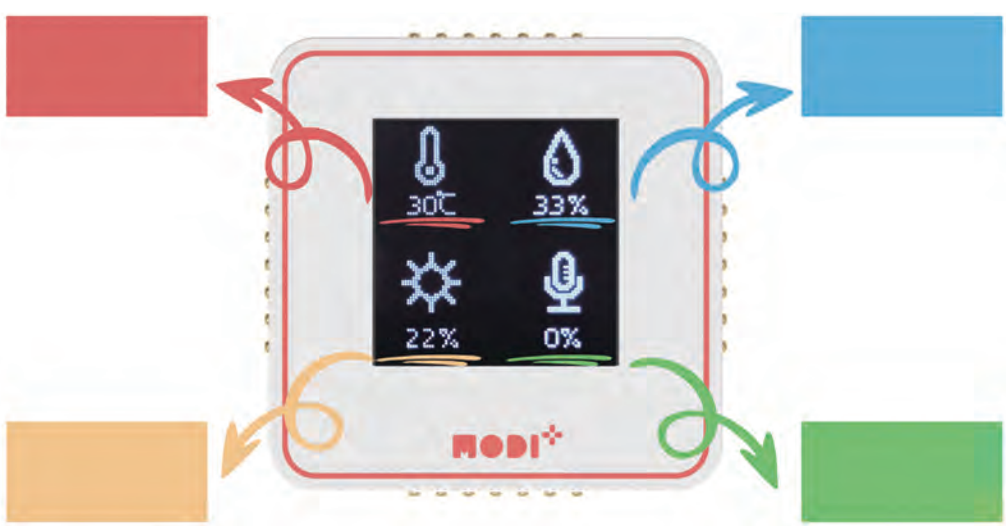
2. **POWERING UP YOUR MODULES** **There are two cool ways to make your modules work:**

a. Use the Network module to connect to your PC.

b. Use the Battery module to power your connected modules.



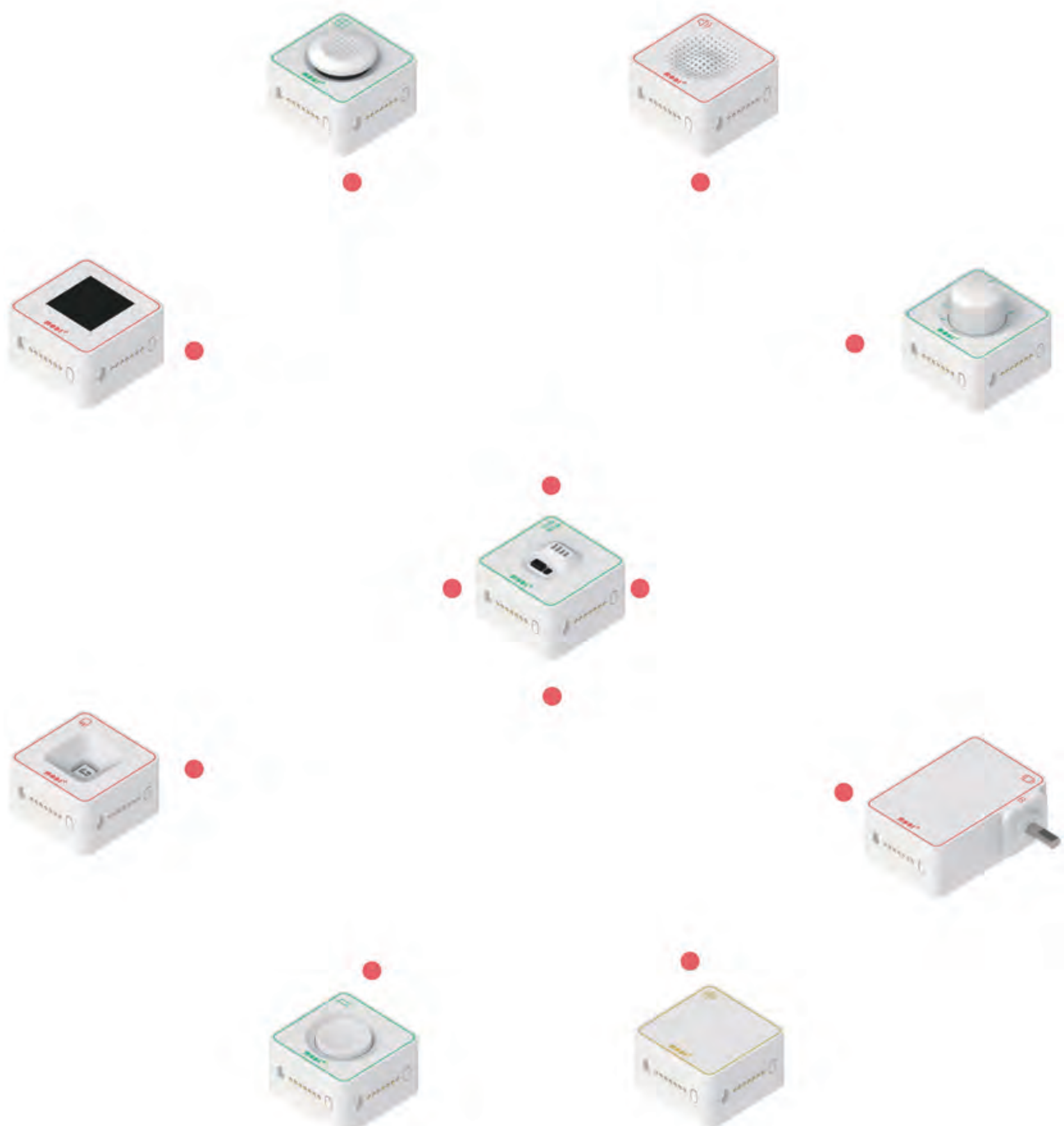
3. **CHECKING THE DISPLAY** Your Display module shows 4 types of information from the Environment module. Identify each one (temperature, humidity, light level, and sound level) and fill in the blanks below.

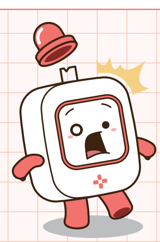


Activity2 Connecting Worlds: Environment and Output

Instructions:

Find 4 MODI modules that can interact with the Environment Module and connect them.



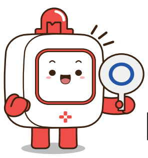


Remember: MODI modules work like parts in robots and computers. Make sure you connect the Environment Module to other modules using an "input" to "output" connection.

UNIT 3 NAVIGATING CODE EDITOR

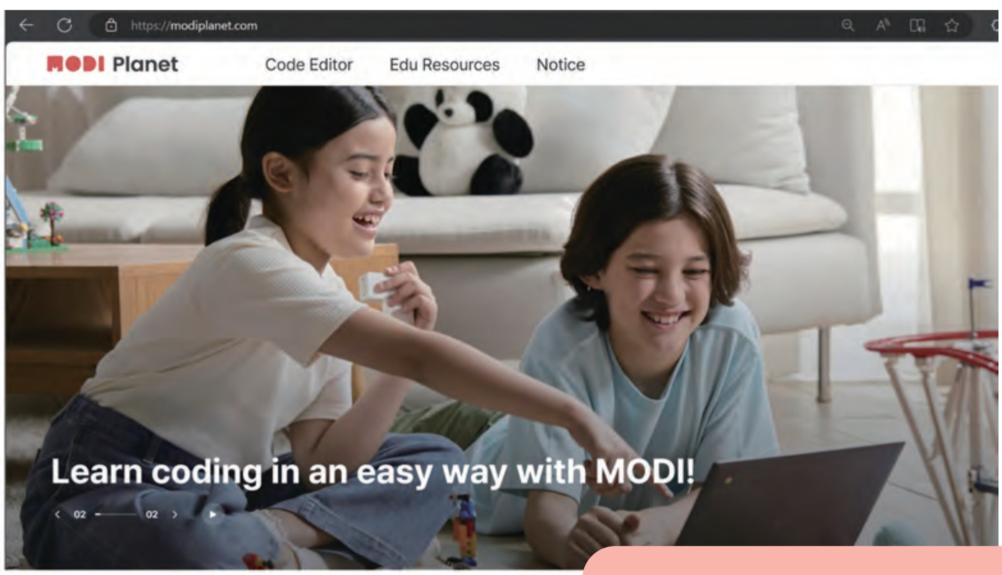
Introduction to Code Editor

What is Code Editor? Code Editor is where we can programm MODI modules with block-based programming.

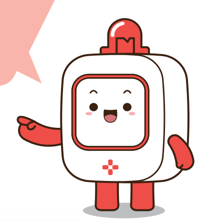


Let's get started by following the next steps:

Step 1 Go to 'MODI Planet' in your web browser. Here's the link: www.modiplanet.com

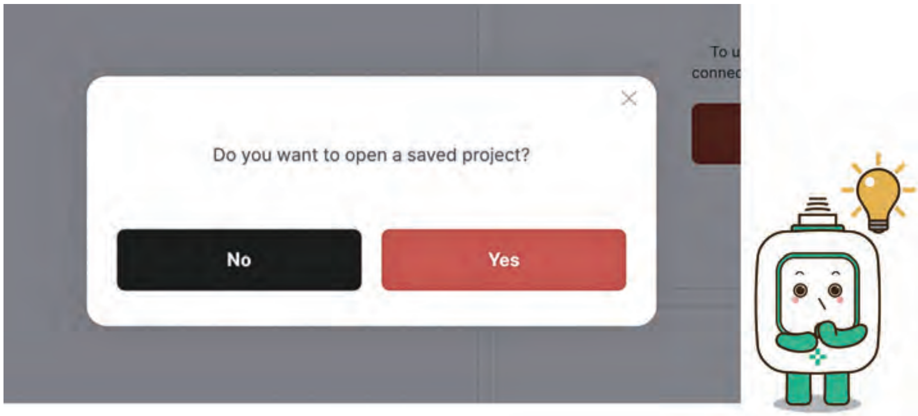


You can also search 'MODI planet' in a search engine!

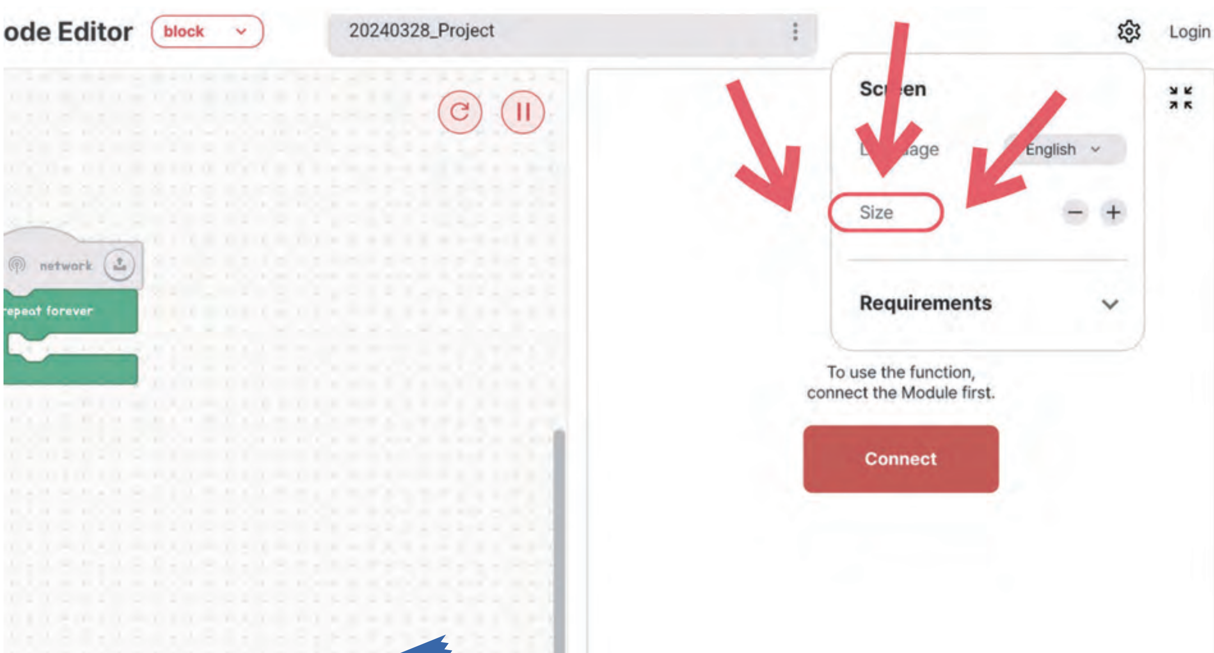


Step 2 Find 'Code Editor' in the top menu and click it.

Step 3 If it asks about opening a saved project, click 'No.'

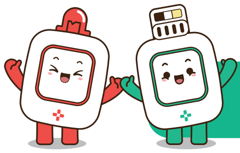


Step 4 If the screen layout doesn't suit your preference, adjust the ratio by clicking the "Set" button.

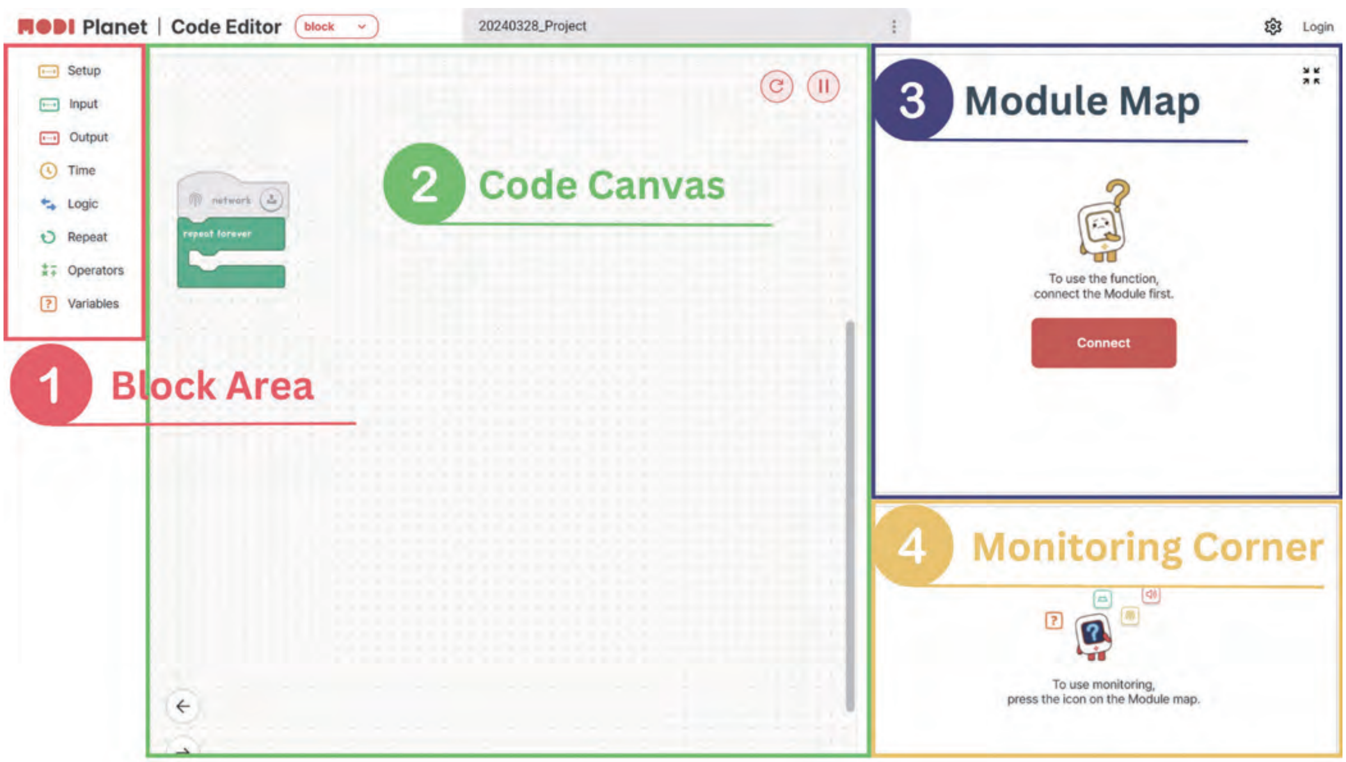


TIPS

Depending on your screen size, some options might look different. If you see a button in the top right corner, use it to see more options.



Exploring Code Editor



❶ Block Area :

This is where all the coding blocks live. They are sorted into eight groups based on what they do.

❷ Code Canvas :

This is your space to build your code by putting blocks together from the Block Area.

❸ Module Map :

Here, you'll see icons for all the MODI modules connected to you PC. Clicking a module icon lets you check on that module.

❹ Monitoring Corner :

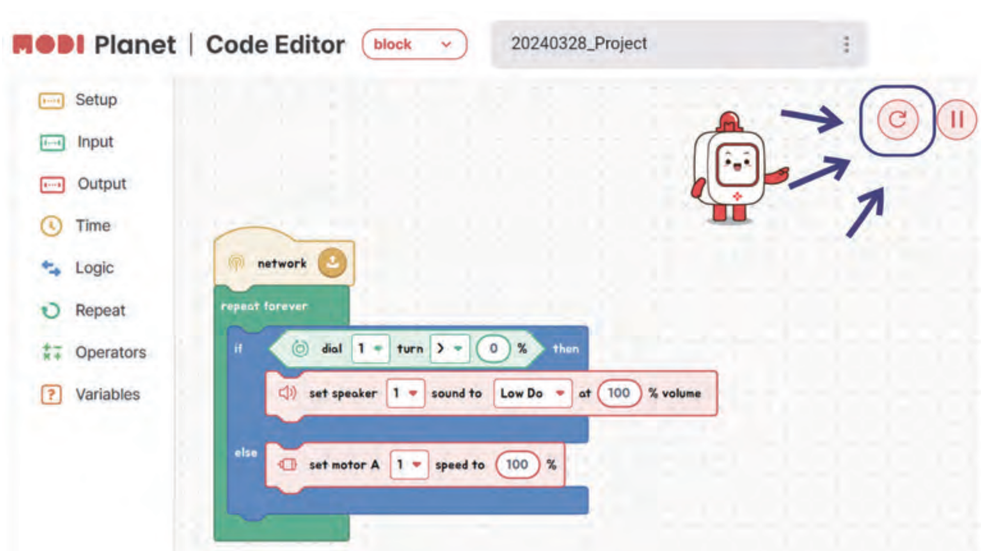
This is where the monitoring screen appears. Depending on the size of your screen, it may appear as a pop-up window.

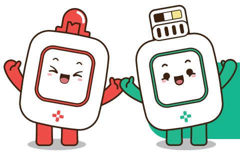
Save & Load Projects

Click the ellipsis in the picture below to open a popup with options to save and load your projects.

Initializing modules

❶ Ensure the modules you wish to initialize are connected to your device.
❷ Click the 'Initialization' button to reset them.





Connecting MODI to device

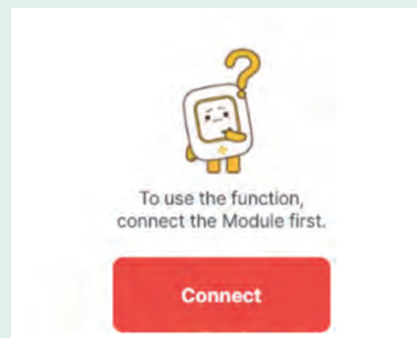
Once the MODI Code Editor is set up, let's connect your MODI modules to your device.

Step 1

Connect the Network Module to your device using a Type-C cable.

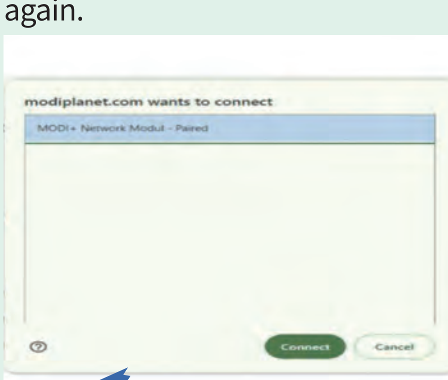
Step 2

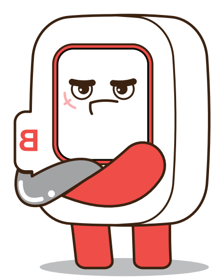
Click the connect button in the module map.



Step 3

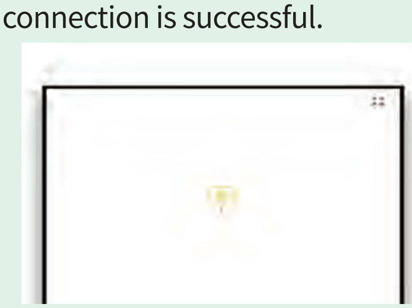
Select 'MODI+ Network Module-Paired' and then click 'Connect' again.

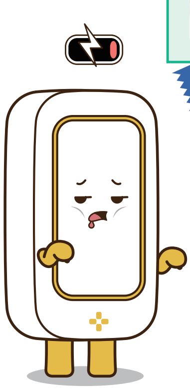




Step 4

A network icon will appear in the Module Map when the connection is successful.





TIPS

Any additional modules connected alongside the Network Module will be displayed in the Module Map.

Types of blocks in the block area

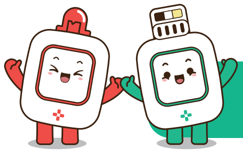
The Block Area is organized into seven tabs, each housing specific types of blocks:

Setup	1	**Network-Related Blocks:** These include the essential code start blocks that initiate your program.
Input	2	**Input Module Blocks:** Tailored for interacting with various input modules.
Output	3	**Output Module Blocks:** Designed to control and utilize output modules.
Time	4	**Time Block:** Used to pause the code execution for a set duration.
Logic	5	**Conditional Statements:** Includes 'if' and 'if-else' blocks for making decisions.
Repeat	6	**Loop Statements:** Contains blocks for repeating actions, like 'repeat'.
Operators	7	**Operators:** Focus on arithmetic calculations, comparisons, and generating random values.
Variables	8	**Variable Blocks:** Used for creating and modifying variables, crucial for storing and using data in your programs.

Blocks: Your First Step in Coding

Think of each block in the Code Editor as a piece of a puzzle that helps you build your program. Just like when you write instructions in a game or a recipe, blocks are the instructions you give to your computer.

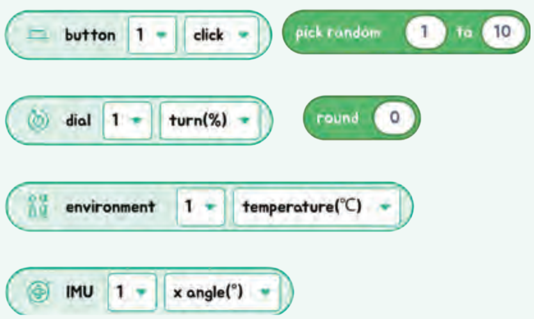
- **Network, Input, and Output Blocks:**
  - These blocks are like saying "go" in your program, telling it to start, take in information, or make something happen!
- **Time, Logic, and Loop Blocks:**
  - These blocks help decide what happens next, like choosing when to jump or keep running in a game.
- **Math and Variable Blocks:**
  - These blocks help you work with numbers and keep track of important data, like scores or clues.



Unplugged

In the Code Editor, there are three different shapes of blocks you'll use to create your programs. Each shape has a special job to do!

1



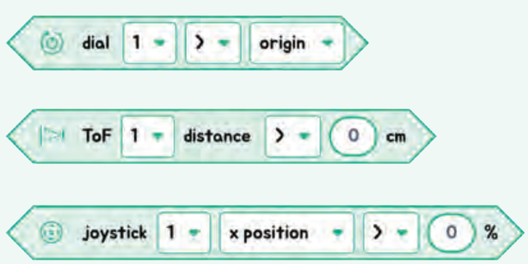
❶ Rounded Blocks:

These blocks hold values, like numbers. Think of them as small baskets carrying important information, usually numbers, for your code.

Examples:

- The button block tells when a button is clicked.
- The environment block can tell the temperature.

2



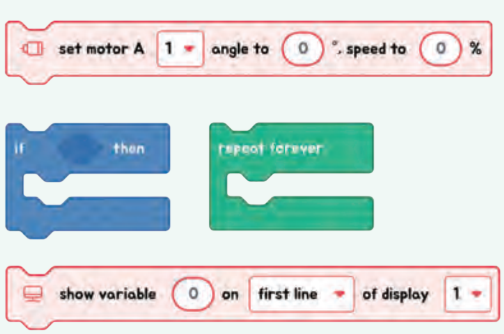
❷ Angled Blocks:

These blocks ask "yes" or "no" questions, dealing with conditions. If the answer is "yes" (true), one thing happens; if the answer is "no" (false), something else might happen.

Examples:

- The distance block asks, "Is this object close?"
- The joystick block asks, "Is the joystick pointing left?"

3



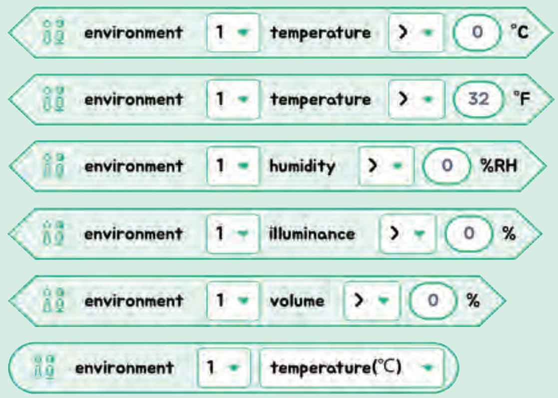
❸ Rectangle-Shaped Blocks:

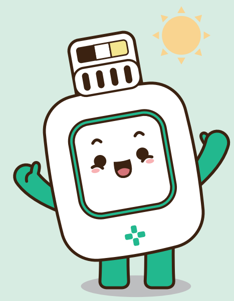
These blocks are all about action. They tell your MODI modules what to do, like moving, lighting up, or making sounds.

Examples:

- The motor block tells a motor to start turning.
- The show variable block displays information on a screen.

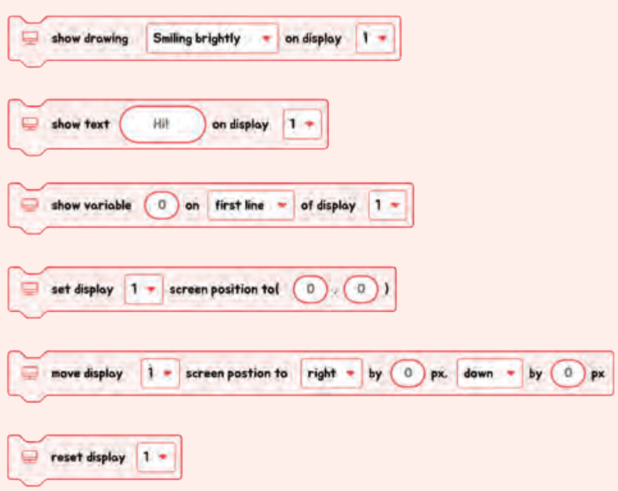
Blocks of Environment

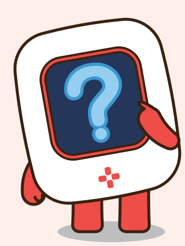




The input module called Environment doesn't perform actions. Instead, it provides 5 condition blocks and 1 value block, each related to different types of environmental data it can sense. The condition blocks are used to check specific environmental conditions, and the value block is used to retrieve the exact measurements of those conditions.

Blocks of Display

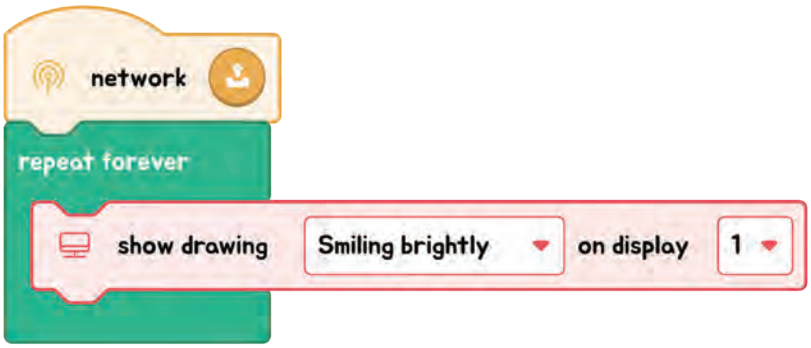




The output module, Display, is all about taking action. It has blocks specifically designed for showing pictures, text, or variable values on the screen. There are also blocks that allow you to move the displayed content around, and one important block that resets or initializes the Display for a fresh start.

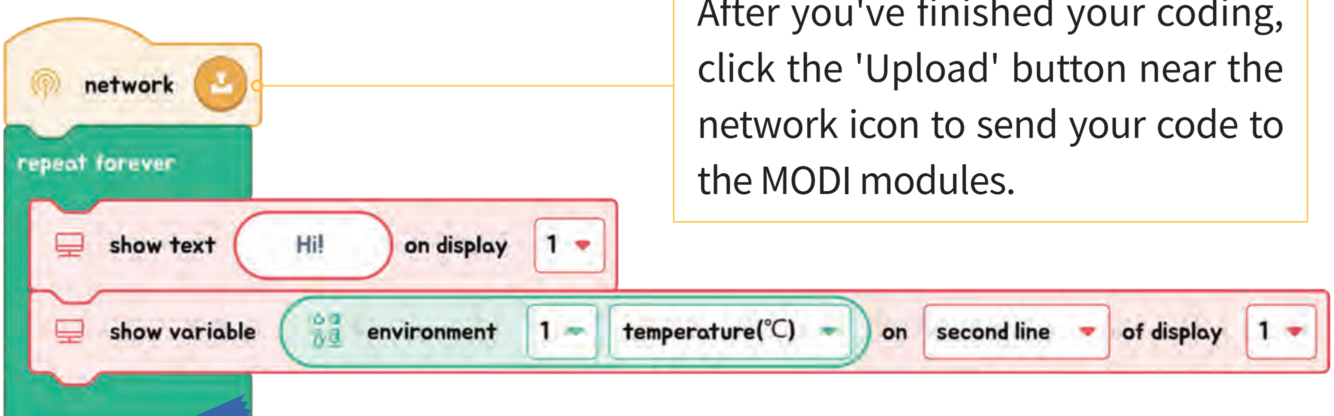
Showing a Picture on the Display

Let's get a picture to appear on the display screen. You can choose a different picture from the list and see how it looks!



Displaying Values on the Display

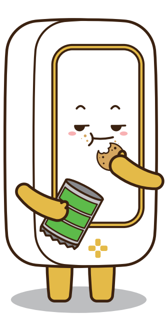
Now, let's have the display screen show the word 'temperature' and then, right below it, show the actual temperature that the Environment module detects.

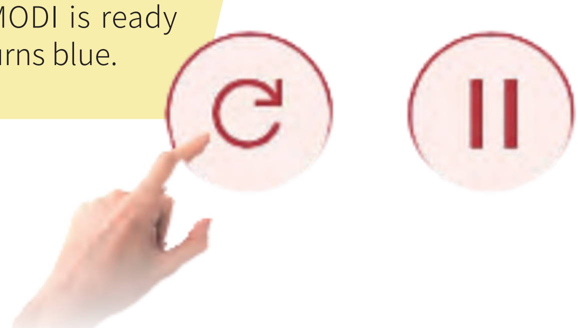


After you've finished your coding, click the 'Upload' button near the network icon to send your code to the MODI modules.

TIPS

To prepare MODI for a new set of code, simply press the 'Initialize' button. You'll know MODI is ready when the status LED on the module turns blue.





CHAPTER 3

CONDITIONAL STATEMENTS

CONTENT

UNIT 1 IF STATEMENT

Description: Learn how to use the "if" statement to make decisions in your code. Understand how this simple control structure allows a program to execute specific actions based on certain conditions. Explore real-life examples and practice writing your own "if" statements to control the flow of your programs.

UNIT 2 IF ELSE STATEMENT

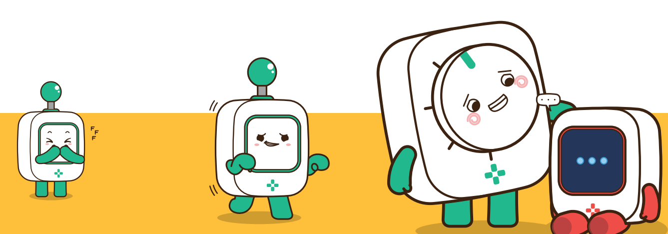
Description: Discover how to extend the "if" statement with the "else" statement to handle multiple scenarios in your code. Learn how to create programs to make decisions with two possible outcomes, improving your ability to handle more complex situations. Practice w

UNIT 3 REPETITION STATEMENTS

Description: Explore the concept of repetition statements, also known as loops, to perform actions multiple times. Understand how loops like "for" and "while" help automate repetitive tasks and simplify your code. Practice creating loops to handle repetitive tasks efficiently and effectively in your programs.

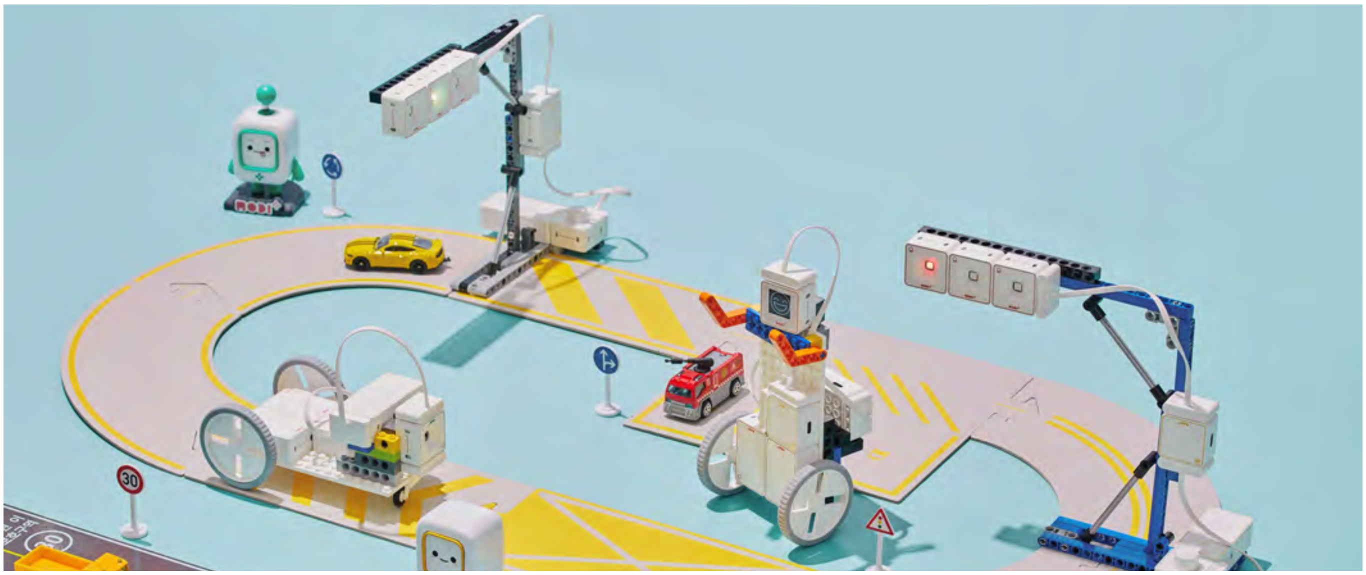
UNIT 4 VARIABLE

Description: Learn about variables and how they store and manage data in your programs. Understand how to create, assign, and use variables to hold different types of information. Explore examples and exercises to see how variables make your code more flexible and dynamic.



CHAPTER 3

CONDITIONAL STATEMENTS



What You Will Learn

- Use "if" statements to make decisions in your code and control the flow based on conditions.
- Extend "if" statements with "if else" to handle multiple outcomes and respond to different scenarios.
- How repetition statements, such as "for" and "while" loops, allow you to perform tasks multiple times and simplify your code.
- The role of variables in programming, including how to create, assign, and use them to store and manage data.

Learning Objectives

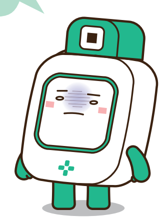
- Understand and apply basic "if" statements to control program execution based on specific conditions.
- Utilize "if else" statements to handle complex decision-making scenarios and manage multiple outcomes in your code.
- Write and use repetition statements (loops) to automate repetitive tasks and improve the efficiency of your code.
- Define and work with variables to store and manipulate data, making your programs more dynamic and flexible.

UNIT 1 IF STATEMENT

Think of an "if statement" like a rule or a condition in a game. It's like saying, "If something is true, then do something."

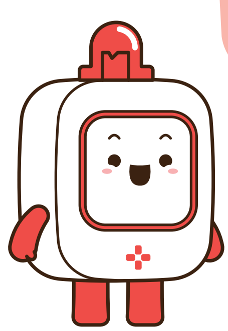
Imagine you have to decide whether to bring an umbrella outside. The rule could be: "If it's raining, take an umbrella. If it's not raining, you don't need to take an umbrella.“

Can you find another example where we can use the "if" statement in our daily life?

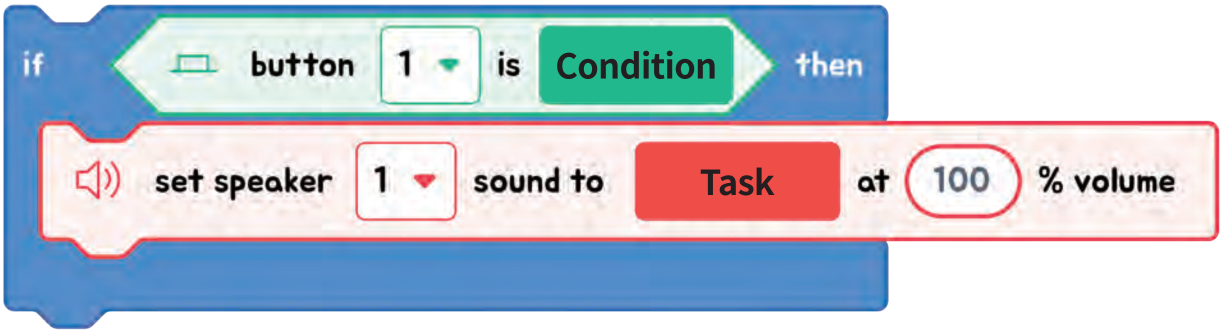


THINK ABOUT IT!

Think of another everyday situation where an "if" rule could apply.
How would you phrase it?



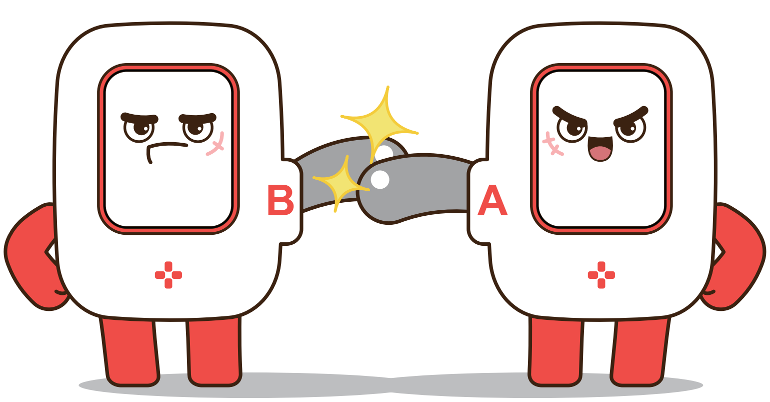
Here's a simple explanation and example:



In block coding, the "if" statement is used to check a condition and perform a specific task only if the condition is true. It allows the program to behave differently based on certain conditions. If you see the "if" statement block, the condition comes between "if" and "then". It checks whether a condition is true, and when it is true, it performs the task.

1 **CONDITION CHECKING** First, the program checks a specific condition. For example, it might check if button is clicked

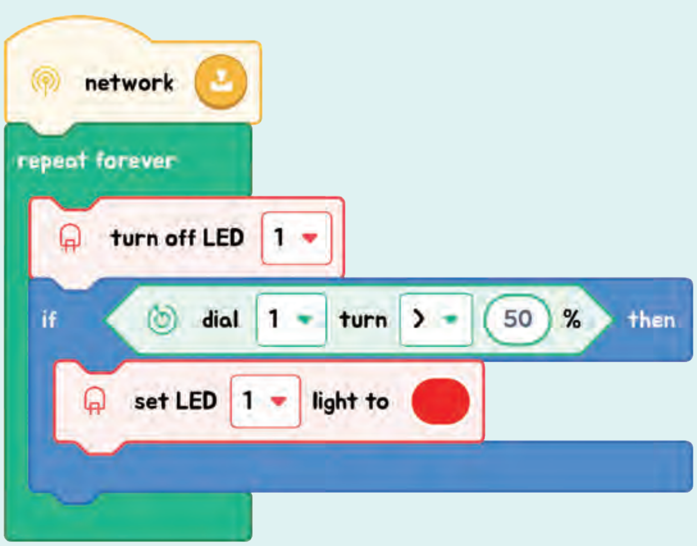
2 **EXECUTION IF TRUE** If the condition is true, the block of code inside the "if" statement is executed. This block contains the tasks that should be performed if the condition is true.



Exercise

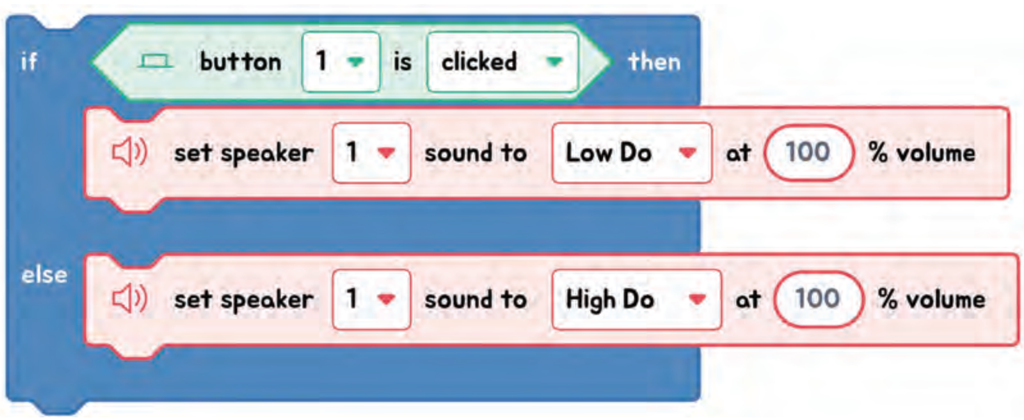
1. In this block example find the condition and task!

2. Read the code and guess what will happen when we turn the Dial less than 50% and over than 50%.



UNIT 2 IF ELSE STATEMENT

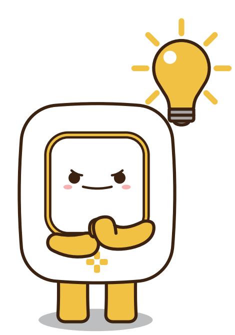
If else statement is similar to the if statement; it just adds an else part, which means when the if condition is false, it follows a task in the else part.



EXECUTION IF FALSE

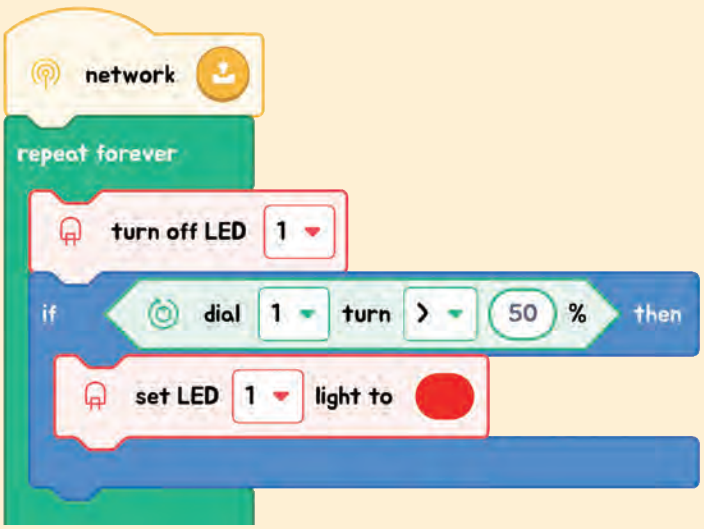
While in most cases only the 'if' block is present, sometimes tasks can be specified to execute if the condition is false. Therefore, in this case, when the button is not clicked, the speaker makes the sound as High Do with 100% volume.

The main point of the if-else statement is to execute the actions specified in the else part if the condition of the if statement is not met!

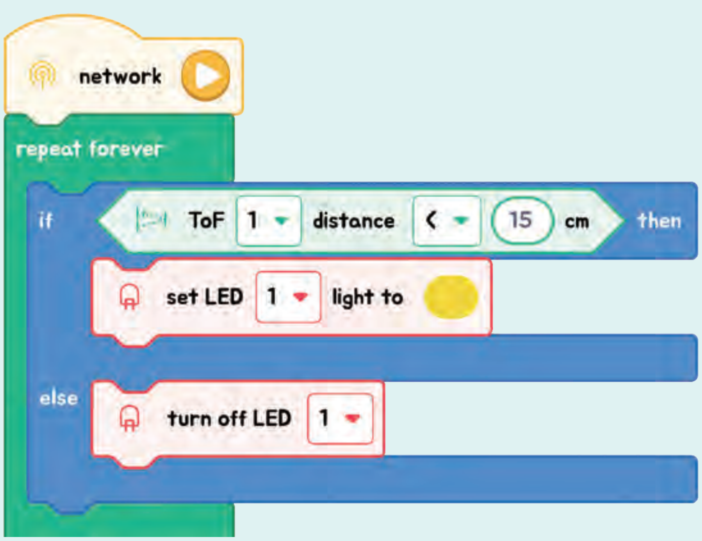


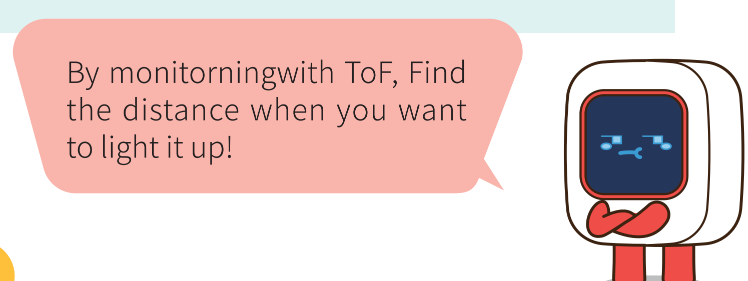
Exercise

1. Change this code to if-else statement in Code Editor.



2. **Make the code in Code editor based on the story below:**
We will make a sensor light, if an object is near by the sensor the light will be on! if not the LED is off.





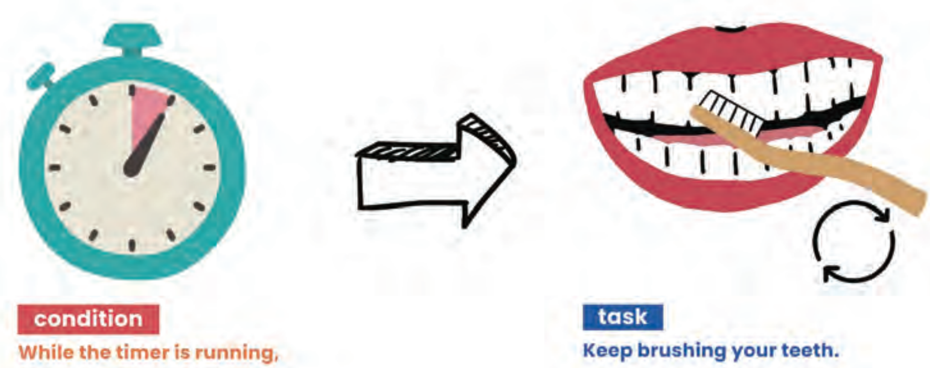
UNIT 3 REPETITION STATEMENTS

WHILE STATEMENT

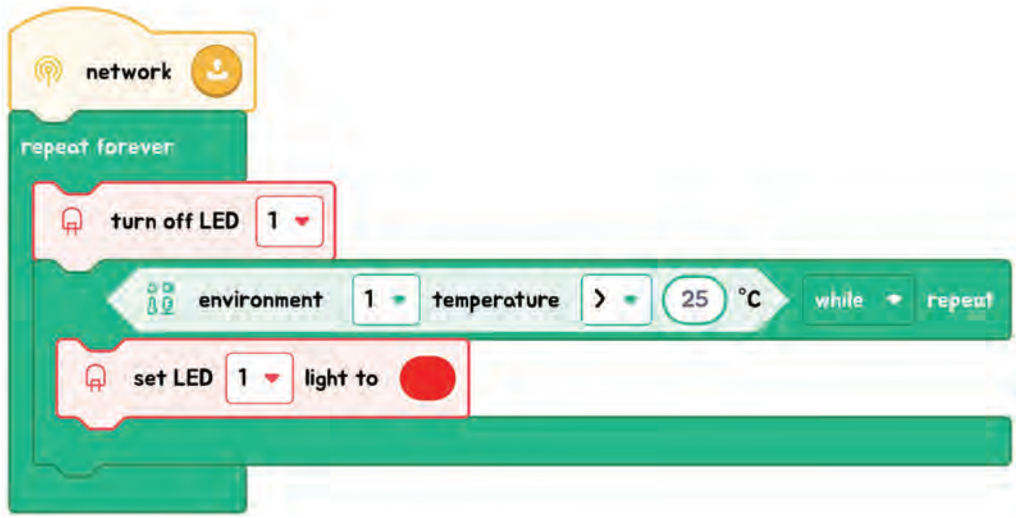
A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing.



EXPLANATION WITH CODE EXAMPLE



In our code example, we're working with a light sensor and an LED. When we start, our LED light is turned off. But we have a special rule written in our code about when to turn it on and make it yellow.

1) Is the illuminance level below 30%?

This checks how bright the light around us is. If it's really dim, less than 30% bright, our rule is met.

2) Set the LED to yellow.

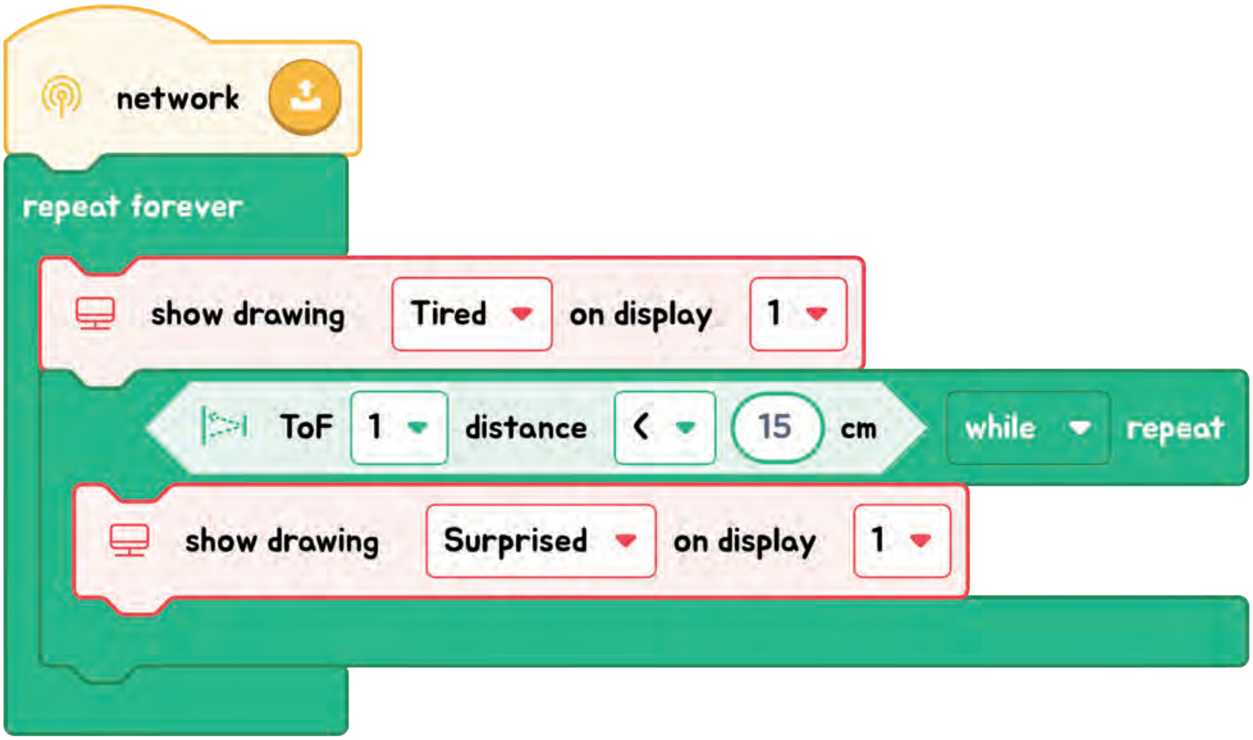
Once we know the light is dim enough, our code tells the LED to turn on and shine in a yellow color. This makes the LED light up yellow as long as the room stays dim.

Code Insight

1. **Default State**: Initially, the LED is off. This state persists until the "**while**" loop starts and when the llluminance condition isn't met.
2. **Condition Checking**: The code first checks if the light level is below 30%, determining the next steps.
3. **Execution While True**: With light levels under 30%, the LED turns yellow, repeating this task as long as the dim condition lasts.
4. **Exiting the Loop**: When light levels exceed 30%, the loop ends, and the LED shuts off, awaiting the next dim condition.

Example

1 **We are spooking a tired MODI, when we are near by the MODI, it's surprised. Find the condition and the action(task) in the code.**



2 **What is the purpose of a while loop in programming?**

Ⓐ To execute a block of code only if a certain condition is true

Ⓑ To repeatedly execute a block of code as long as a certain condition is true

Ⓒ To execute a block of code regardless of any condition

Ⓓ To execute a block of code once and then exit the loop

CORRECT ANSWER Ⓑ

3 **When does a while loop stop executing?**

Ⓐ When the condition inside the loop is false

Ⓑ When the condition inside the loop is true

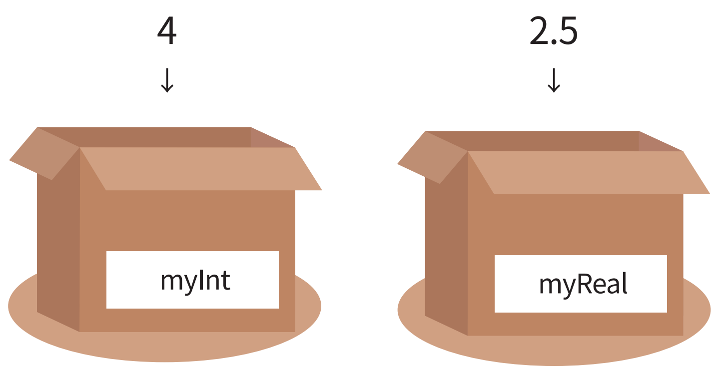
Ⓒ When the loop encounters an error

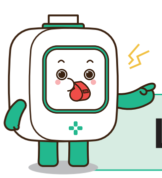
Ⓓ When the loop reaches a specific line number

CORRECT ANSWER Ⓐ

UNIT 4 VARIABLE

A programming variable is a named storage location in a computer's memory that can hold data, such as numbers, or other values. Variables are used to store and manipulate data within a program. They allow programmers to work with data dynamically, enabling them to perform calculations, make decisions, and control the flow of a program.

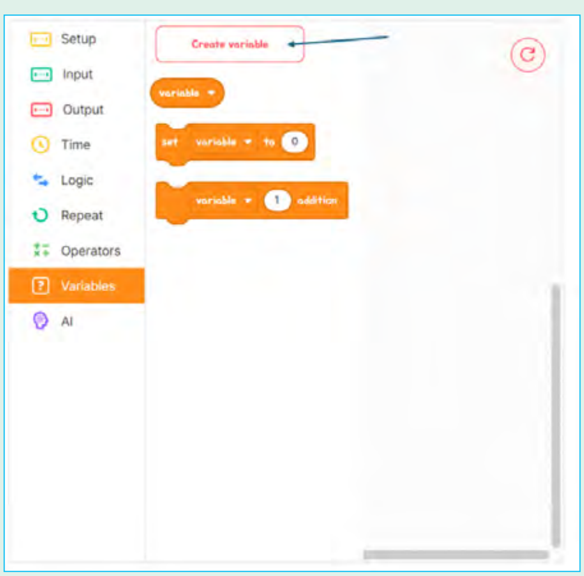


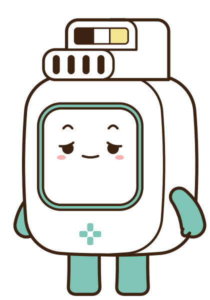


Let's make the variable in Code Editor!

Step1

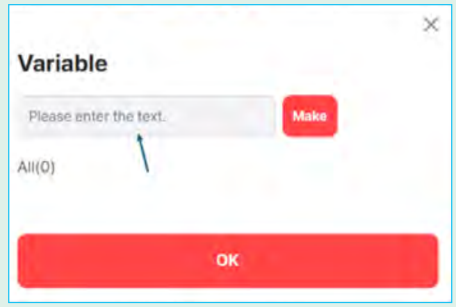
Click the Variables section and **Create variable**

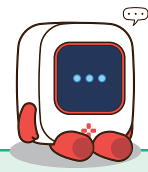


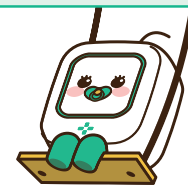


Step2

Tpye the variable name (mode).





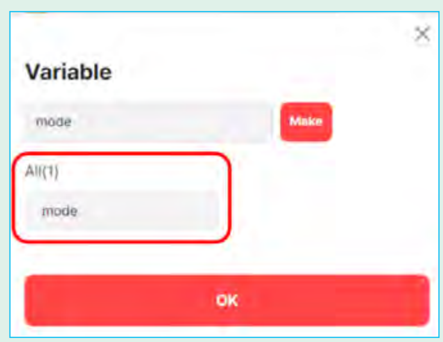


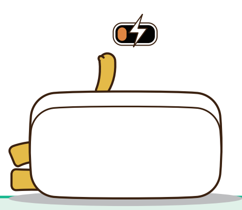
Step3

After type the name of the variable and **Make** click the.

Step4

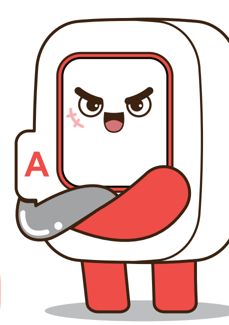
After step3, you can see the variable that you made on the list.





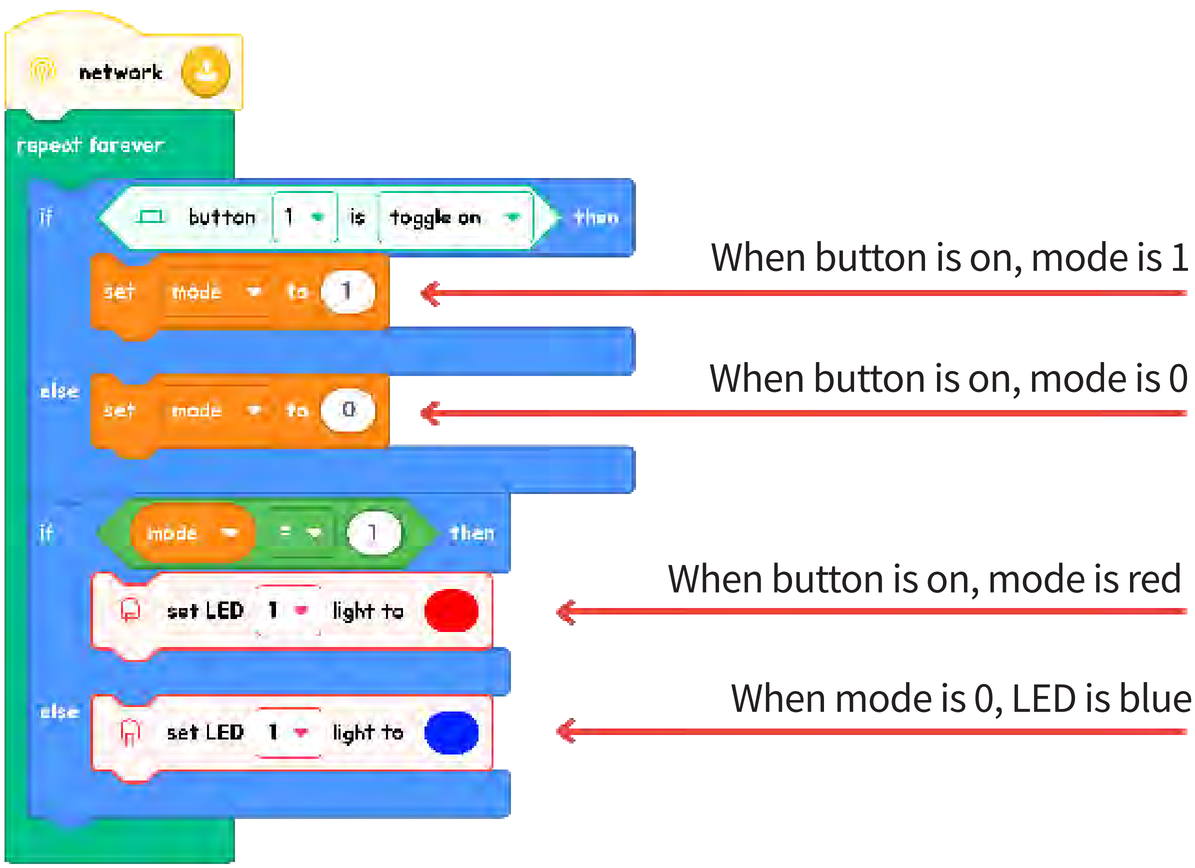
Step5

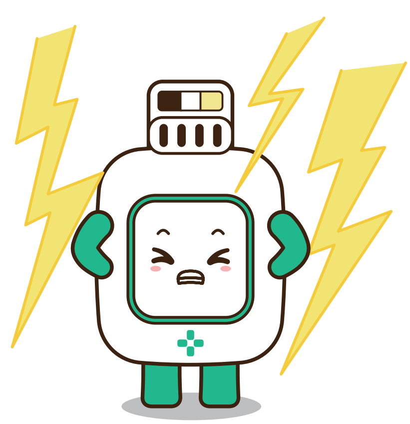
When you click on the variable scroll part, you can see the created variables.



Let's see the example with variable 'mode'.

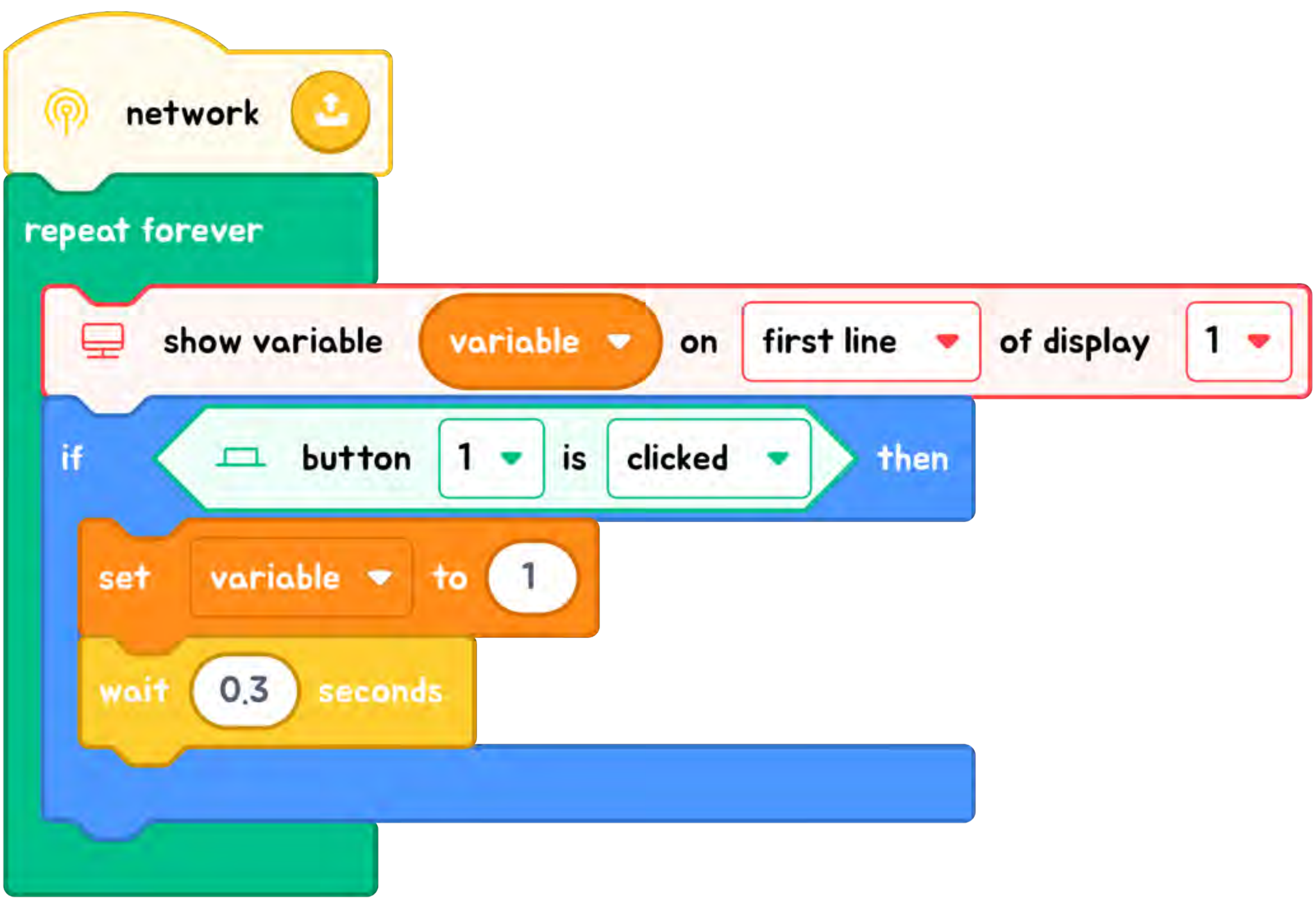
The color of the light changes depending on the on/off status of the button. When the mode value changes, the color of the light changes accordingly. The mode is initially set to 0, so the light appears blue. When you press the button, the mode changes to 1, and you can see the light turning red.



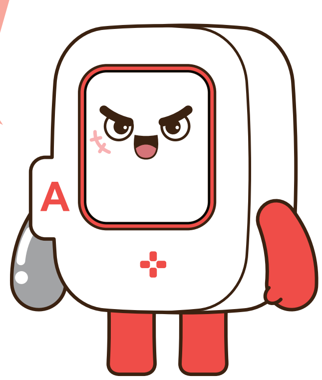


Example

1. Every time the button is clicked, we want to increment the count of clicks (variable). With the addition block, we can easily track the result. In block coding, when a variable is created, its initial value is 0 unless specified otherwise.



If you want the number to decrease every time the button is clicked, you should change it to subtraction operation. So, you need to modify the code to perform subtraction operation.



CHAPTER 4

CARBACKUP SENSOR

CONTENT

UNIT 1 WHAT IS A SENSOR?

Description: Explore the fascinating world of sensors and discover how they work. Sensors are devices that detect environmental changes and provide information about them. Learn about the different types of sensors and how they are used to gather data in various applications.

UNIT 2 WHAT IS A BACKUP SENSOR?

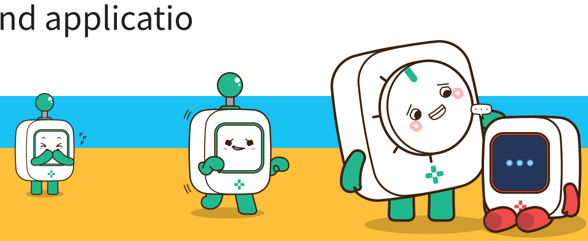
Description: Delve into the specific type of sensor known as a backup sensor. This unit explains how backup sensors help drivers detect obstacles behind a vehicle, making reversing safer. Understand the technology behind these sensors and their importance in preventing accidents.

UNIT 3 CREATING A CAR BACKUP SENSOR

Description: Get hands-on experience by building your own car backup sensor. This unit guides you through creating a simple backup sensor system using basic components. Learn how to assemble and program the sensor to alert you when objects are detected behind a vehicle.

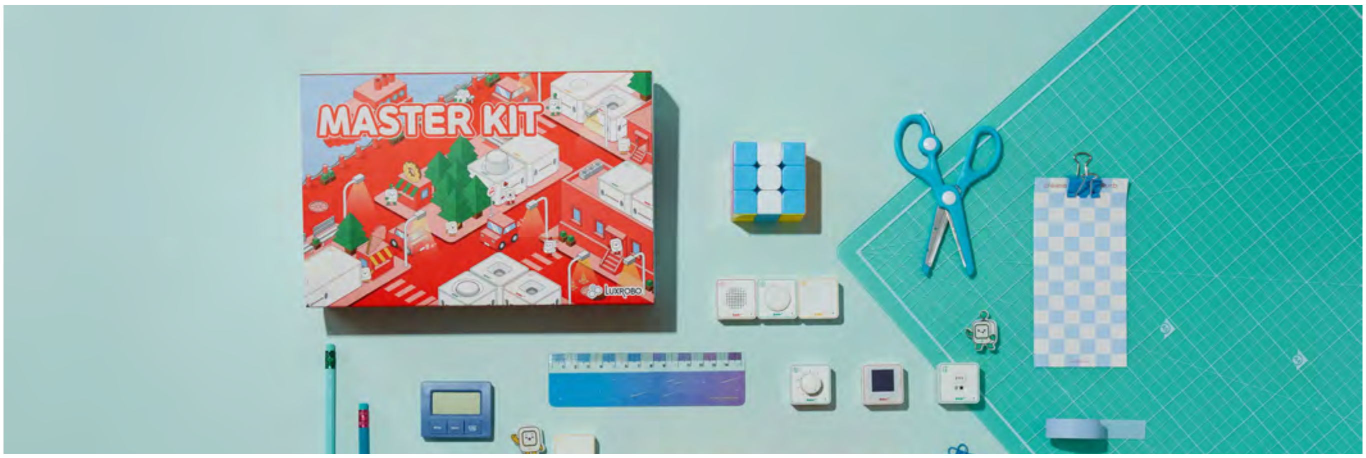
UNIT 4 DIVE INTO FLOW

Description: Explore how sensors are used in everyday life beyond car backup systems. This unit provides examples of sensors in common objects and applicatio



CHAPTER 4

CARBACKUP SENSOR



What You Will Learn

- **How Sensors Work:** Discover the fundamental principles behind sensors and their ability to detect and respond to environmental changes.
- **Backup Sensor Functionality:** Understand the specific role of backup sensors in vehicles, including how they help prevent accidents by detecting obstacles.
- **Creating a Sensor System:** Learn to build and program a basic car backup sensor, gaining hands-on experience with sensor technology.
- **Everyday Use of Sensors:** Explore how sensors are integrated into various everyday items and applications, and appreciate their impact on daily life.

Learning Objectives

- **Identify Sensor Types:** Recognize different types of sensors and describe their functions in detecting environmental changes.
- **Explain Backup Sensor Mechanisms:** Understand and articulate how backup sensors work to enhance safety while reversing a vehicle.
- **Build a Backup Sensor System:** Demonstrate the ability to create and program a simple car backup sensor using basic electronic components.
- **Apply Sensor Knowledge:** Apply knowledge of sensors to identify their uses and benefits in various everyday situations.

UNIT 1 WHAT IS A SENSOR?

Have you ever seen automatic doors that open when you walk near them or lights that turn on themselves when you enter a room? These work because of a sensor!

A sensor is a special device that can sense or "feel" what is happening around it. It can detect light, heat, movement, or even sound. Once a sensor detects something, it sends a signal to a machine, telling it to take action. For example, a sensor on an automatic door detects when a person is nearby and signals the door to open.

Types of Sensors:


1 **MOTION SENSORS** These detect movement. For example, a motion sensor might turn on a light when you walk in front of a motion sensor.

2 **LIGHT SENSORS** These detect how much light is in an area. Streetlights often use light sensors to turn on when it gets dark outside.

3 **TEMPERATURE SENSORS** These can sense how hot or cold it is. Your home thermostat uses a temperature sensor to control your heating or cooling system.

HOW DO SENSORS WORK?

Let's break it down in a simple way:

1 **DETECTING SOMETHING** The sensor watches for changes in its surroundings (like movement, light, or temperature).

2 **SENDING A SIGNAL** Once it detects something, it sends a message (signal) to a machine or device.

3 **TAKING ACTION** The machine takes action based on the signal. For example, it could turn on a light, open a door, or start cooling a room.

Activity: Be the Sensor!

Find the sensors which is in your body! Write the name of the sensor and what it can do.

Sensor	What does it do?
Vision	*Get visual information..*

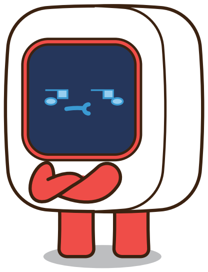
UNIT 2 WHAT IS A BACKUP SENSOR?

Have you ever wondered how cars can "see" what's behind them? Imagine if you were driving backward but couldn't see what was behind you. That sounds a little scary, right? This is where a special tool called a backup sensor helps!

WHAT IS A BACKUP SENSOR?

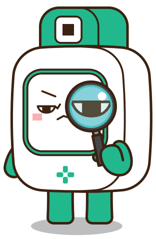
A backup sensor is a helpful tool that cars use to keep drivers safe. It helps the driver know if there's something behind the car while backing up.

For example, if you are reversing and there's a wall behind the car, the sensor can "sense" the wall and let the driver know by making a beeping sound or showing lights on the car's dashboard.



HOW DOES IT WORK?

Here's how a backup sensor helps:



1 **Detects Objects :** When these waves hit something behind the car, like a bicycle or a tree, they bounce back to the sensor.

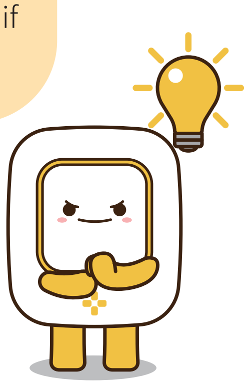
2 **Alerts the Driver :** The car will beep or flash a light to warn the driver to stop, preventing a crash.

WHAT HAPPENS WHEN YOU GET CLOSE TO AN OBJECT?

The backup sensor is clever! If the car is far away, the beeping is slow. But as the car gets closer to an object, the sensor makes the beeping sound faster, helping the driver know when to stop.

Why is it Important?

Backup sensors are like an extra set of eyes. Even when you check the mirrors, you might miss something behind the car. The sensor ensures the driver knows what's there, even if it's hard to see!

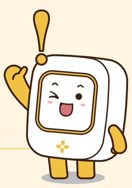


Activity: Finding Sensors in Everyday Life

You will explore their surroundings to identify different types of sensors and understand how they are used in daily life. Look around your home, school, or neighborhood and find at least three sensors.

- Write down:

- Where you found the sensor.
- What it does (e.g., turns on lights, opens doors, detects motion).



UNIT 3 CREATING A CAR BACKUP SENSOR

This section will explore creating a simple alert system using the ToF(Time-of-Flight) sensor and the MODI Speaker Module. This system will help us detect when an object gets too close and sound an alarm when needed.

HOW IT WORKS IN ACTION:

OBJECT CLOSE When an object gets within 20 centimeters of the ToF sensor, the sensor sends a signal to the MODI system. The system instructs the speaker module to sound an alarm, which helps us know that something is too close.

OBJECT FAR If the object moves farther than 20 centimeters, the ToF sensor will no longer trigger the alarm, and the sound will turn off. This indicates that the object is now at a safe distance.

Let's look detail about the Speaker module.

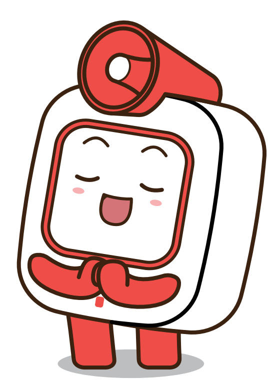
The MODI Speaker Module is a special tool for creating and playing sounds. It's like a little speaker that can make different noises, play melodies, and even perform songs. Let's explore how this cool module works!

The MODI Speaker Module is designed to produce sound. It acts as an output module, which means it takes signals and turns them into sounds we can hear.

When connected to a MODI system, it can play different types of sounds based on your commands.

SINGLE NOTES The speaker module has blocks that can play one note at a time. You can use these blocks to create simple melodies or play individual notes.

MUSICAL BLOCKS Special blocks can also play pre-set tunes or songs. You need to choose the right block, and the module will play the music.

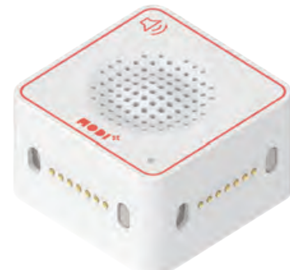


Activity: Create Restroom Indicator

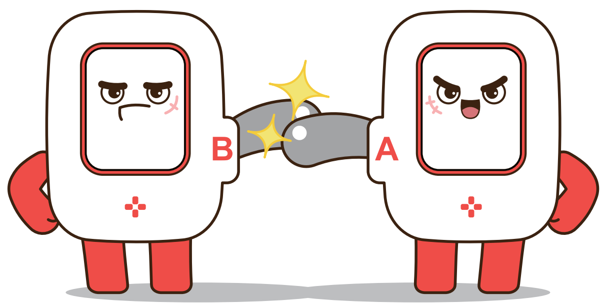
1 Materials Needed:

Network

ToF



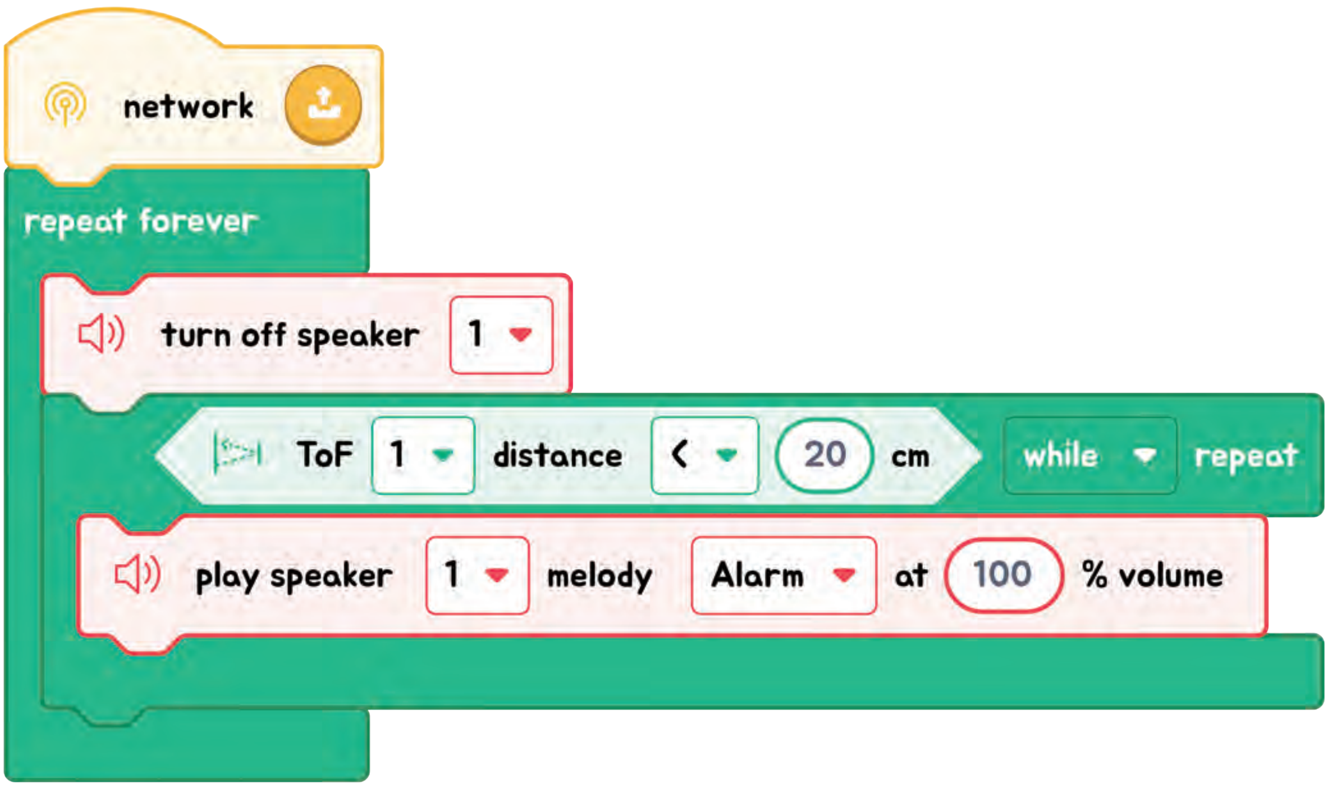
Speaker



2 Code story

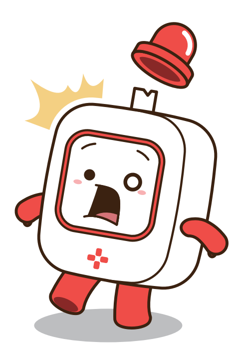
If Dia turn is over than **50%**	
↓ true play an **alarm**	↓ false turn off the **speaker**

Code Quest



Checkpoint

If we want to add LED when object is near by sensor we want to show red light, otherwise the LED is off. Where should we put these blocks?

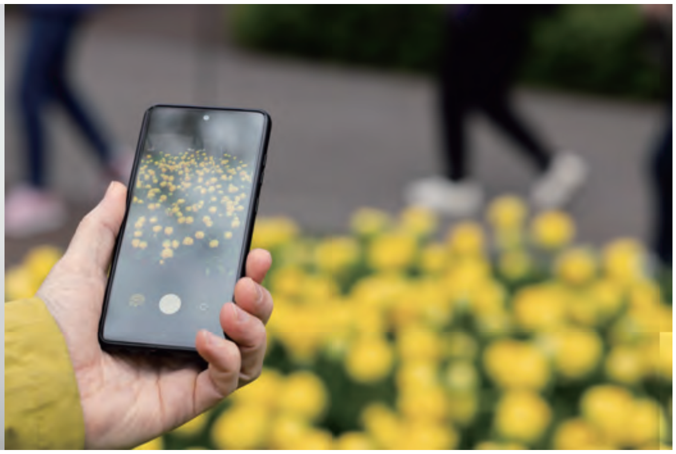


UNIT 4 DIVE INTO FLOW

SENSORS IN EVERYDAY LIFE

Sensors are all around us, helping to make our lives easier and safer. They seem like magical devices, but they're based on simple ideas that help us understand the world better. This section will explore how sensors are used daily and why they are essential.

Sensors in Action:



AUTOMATIC LIGHTS

How They Work: Motion sensors detect when someone enters a room. When movement is detected, the sensor sends a signal to turn on the light.
Why It's Useful: This saves energy because the lights only turn on when someone is in the room and turn off when no one is there.

THERMOSTATS

How They Work: Temperature sensors measure the room's temperature. The thermostat then adjusts the heating or cooling system to keep the room at the right temperature.

Why It's Useful: It helps keep our homes comfortable by ensuring the temperature is right.

THERMOSTATS

How They Work: Temperature sensors measure the room's temperature. The thermostat then adjusts the heating or cooling system to keep the room at the right temperature.

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SMARTPHONES

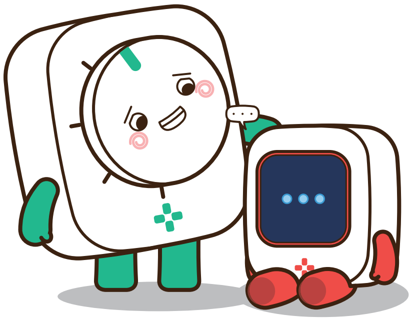
How They Work: Smartphones have several sensors, such as accelerometers that detect movement and proximity sensors that detect when you're holding the phone close to your ear.

Why It's Useful: These sensors help the phone adjust its screen brightness or turn off the display during calls to save battery.

CARS

How They Work: Modern cars have sensors that help with parking, like backup sensors that beep when you're getting close to something behind you.

Why It's Useful: They help drivers park safely and avoid collisions.



Multiple Choice Questions

1 What is the main purpose of a car backup sensor?

Ⓐ To measure the speed of the car
Ⓑ To detect obstacles behind the car
Ⓒ To check the fuel level
Ⓓ To control the car's air conditioning

CORRECT ANSWER Ⓑ

2 Which module can sense the distance?

Ⓐ ToF
Ⓑ Button
Ⓒ IMU
Ⓓ Speaker

CORRECT ANSWER Ⓐ

3 What is a common feature of backup sensors when they detect an obstacle?

Ⓐ The car's engine automatically turns off.
Ⓑ A warning sound or alert is activated.
Ⓒ The car's headlights turn on.
Ⓓ The car's radio volume increases.

CORRECT ANSWER Ⓑ

Reflection Questions

1 How do sensors help us in our everyday lives?

2 Can you think of other devices at home or in public that use sensors?How do sensors help us in our everyday lives?

3 Can you think of other devices at home or in public that use sensors?

CHAPTER 5

MOOD LIGHT WITH IOT

CONTENT

UNIT 1 WHAT IS IOT (INTERNET OF THINGS)?

Description: Discover how everyday objects can connect to the internet and communicate with each other. Learn how IoT makes devices more intelligent, and see how it can be used in your life.

UNIT 2 THE IMPORTANCE OF LIGHT IN EVERYDAY LIFE

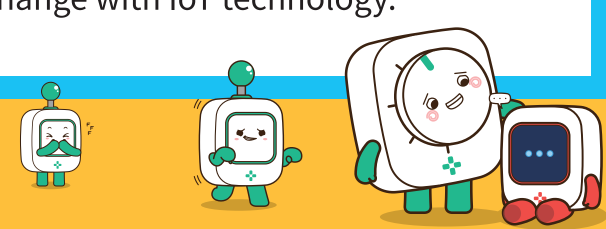
Description: Understand why light is important in different parts of your day. Explore how light helps us see, work, and feel comfortable and why different situations need different kinds of light.

UNIT 3 CREATING YOUR OWN MOOD LIGHT

Description: Build your mood light using simple technology. Learn how to change the colors and brightness to match your mood. See how you can use coding to control the light in fun ways.

UNIT 4 DIVE INTO FLOW

Description: Explore the fascinating journey of light, from the early use of candles to modern-day bright lighting. Learn how advancements in light technology have shaped how we live, work, and play. Discover how light has evolved and how it continues to change with IoT technology.



CHAPTER 5

MOOD LIGHT WITH IOT

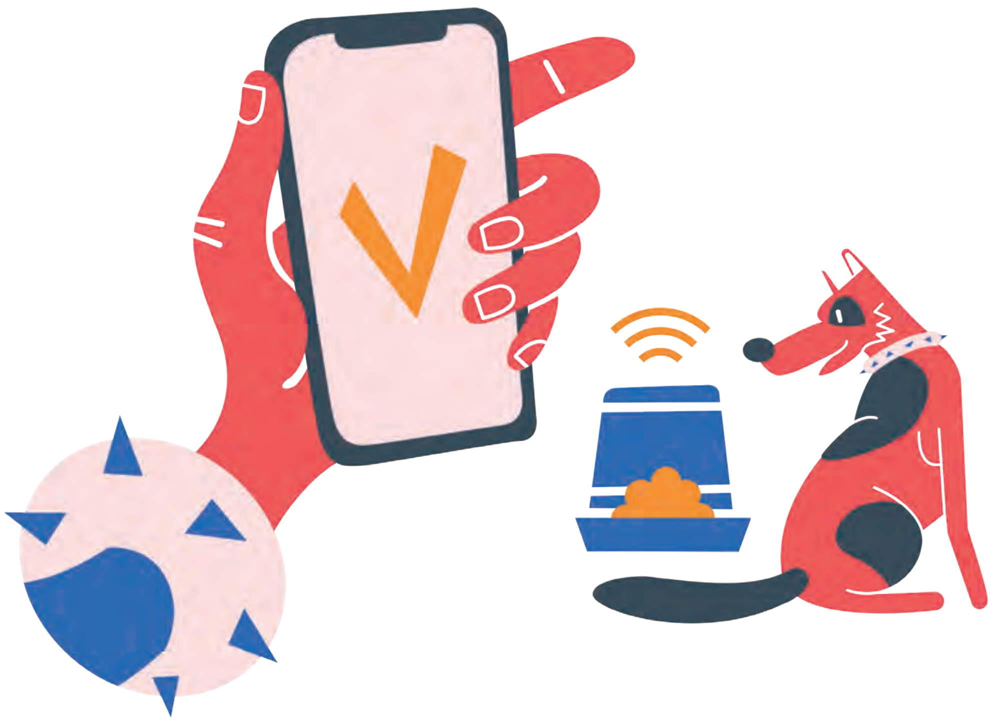
What You Will Learn

- What IoT (Internet of Things) is and how it connects devices to make everyday life more convenient.
- How light is used in everyday life and why different types of lighting are essential for various situations.
- How to create your mood light and control its brightness and color using IoT technology.
- The history and evolution of lighting technology, from early methods to modern intelligent lighting systems.

Learning Objectives

- Understand the concept of IoT and its applications in daily life through connected devices.
- Recognize the crucial role of light in everyday life and the necessity of different lighting types to suit specific environments, enhancing comfort and productivity.
- Learn how to build and control a mood light using IoT and understand how innovative technology can enhance lighting experiences.
- Explore the history of light and lighting technology and understand the role of electricity and intelligent systems in advancing light.

UNIT 1 WHAT IS IOT (INTERNET OF THINGS)?



The Internet of Things, or IoT, is the idea that everyday objects can connect to the internet and communicate with each other. This helps us automate and control devices easily, even from far away. Think of smart lights, thermostats, or even fridges that can send you alerts.

HOW IOT WORKS:

CONNECTING DEVICES

Devices like lights, sensors, and alarms can connect to the internet using Wi-Fi or Bluetooth.

SENDING AND RECEIVING DATA

These devices can send data, like how bright a light is, to your phone. They can also receive instructions, like when to turn on or off.

AUTOMATION

IoT allows us to automate tasks. For example, your mood light can change color automatically depending on the time of day, creating a relaxing atmosphere at night or a bright environment

EXAMPLES OF IOT IN DAILY LIFE:

SMART LIGHTS

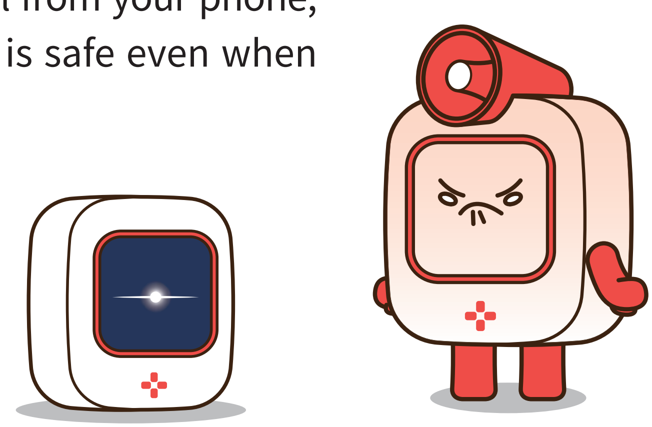
Lights that can change color and brightness based on commands from your phone.

SMART THERMOSTATS

Devices that adjust the temperature automatically depending on the weather or time of day.

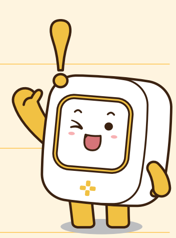
SMART DOOR LOCKS

Locks you can control from your phone, ensuring your home is safe even when you're not there.



Activity: Exploring IoT Devices

Learn how IoT devices work and how they can be controlled remotely. Discuss about IoT: Start by talking about what IoT is and give examples of IoT devices that students may have seen or used before (smart speakers, smart lights, etc.).

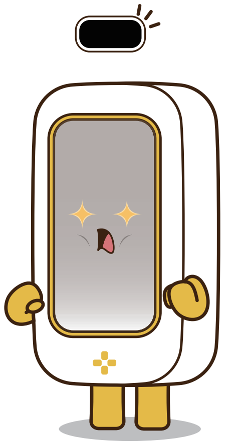


UNIT 2 THE IMPORTANCE OF LIGHT IN EVERYDAY LIFE

WHY DO WE NEED LIGHT?

Light plays an essential role in our daily lives. It helps us see, work, and stay safe. Without light, we would have trouble completing tasks, moving around, or even staying alert.

Different Lights for Different Situations:



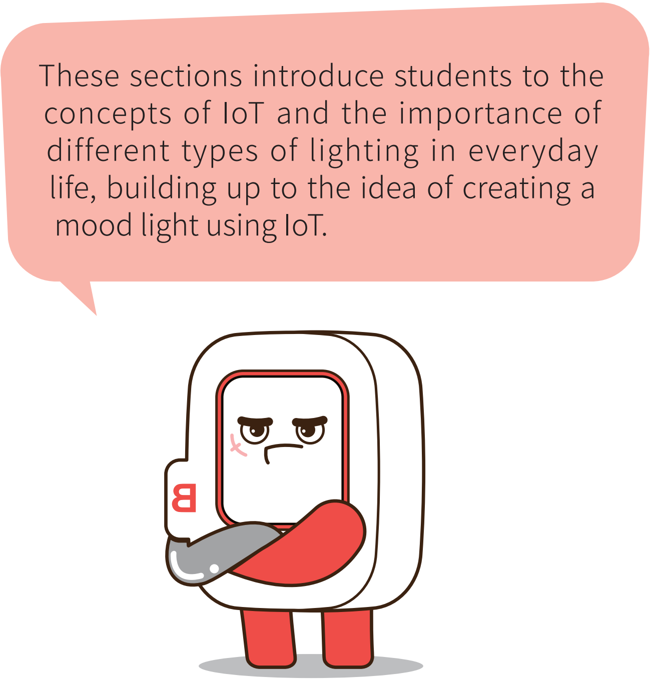
BRIGHT LIGHT FOR WORK When you're studying, reading, or working, bright light helps you stay focused and see clearly. It keeps you awake and makes the environment more comfortable.

SOFT LIGHT FOR RELAXATION In the evening or when you want to relax, softer, dimmer light helps create a calm atmosphere. It makes you feel comfortable and helps your mind unwind before sleep.

COLORFUL LIGHTS FOR FUN OR DECORATION Lights aren't just functional—they can be fun too! Color-changing lights are often used to set a mood for parties, holidays, or even as a way to make your room feel cozy.

How IoT Helps with Lighting:

- With IoT, you can adjust the brightness and color of your lights based on what you're doing. You can create a reading environment with bright white light or a relaxing space with soft blue or warm yellow light—all controlled from your phone.



Activity: Different Lights for Different Moods

Understand how different lighting settings affect mood and tasks.

Instructions:

1 **EXPLORE DIFFERENT LIGHTS**

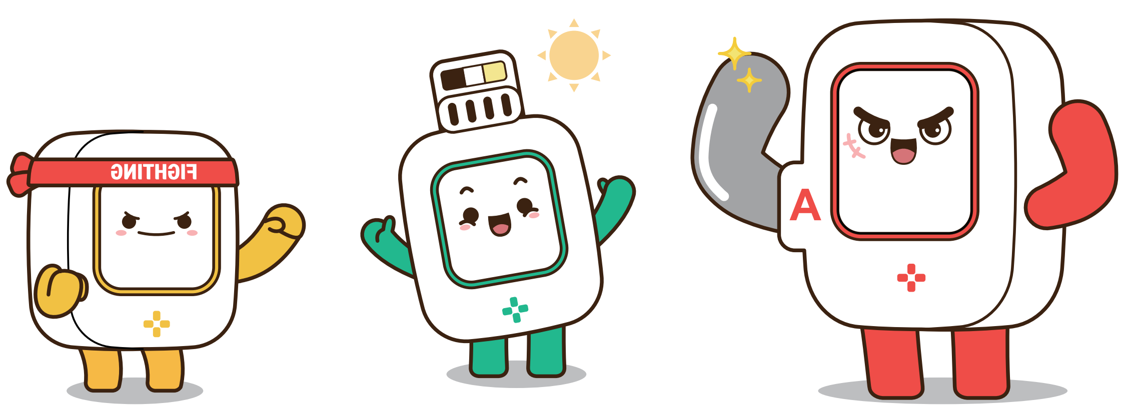
- Turn the light on and show the class how changing the brightness affects the room.
- Use different colored filters or a smart light to display different colors.

2 **CREATE A SCENARIO**

- Ask the students to imagine different scenarios, such as doing homework, reading before bed, or having a party.
- Change the light settings to match each scenario (e.g., bright light for homework, dim light for reading, colorful light for a party).

3 **MOOD REFLECTION**

- After each scenario, ask the students how the lighting made them feel and which light they would prefer in that situation.



UNIT 3 CREATING YOUR OWN MOOD LIGHT

Today, we are going to create a fun Mood Lantern! This lantern changes colors every time you press a button. Let's see how it works:

HOW IT WORKS IN ACTION:

1 Starting with Mode 0 (Off)

When you turn on the lantern, it starts in Mode 0. This means that the light is off, and there is no color yet.

2 Pressing the Button

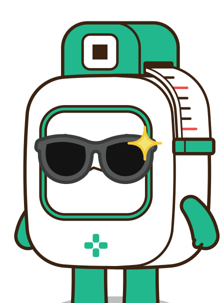
When you press the button, the mode changes. Each mode will make the lantern light up in a different color.

- **Mode 1:** When the mode is 1, the lantern glows yellow.
- **Mode 2:** When the mode is 2, the lantern changes to green.

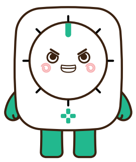
Every time you press the button, the lantern switches between being off, yellow, and green, depending on the mode. This way, you can easily control the mood in the room with just a few button presses!

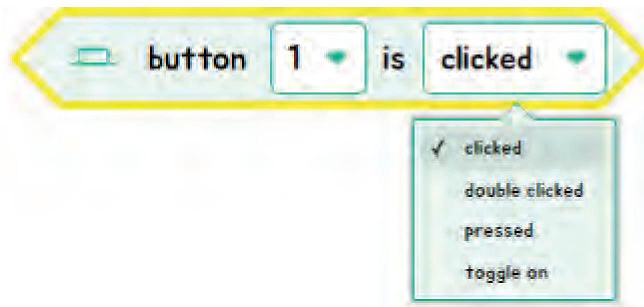
3 Resetting the Mode

If you press the button again and the mode goes above 2, it will reset back to Mode 0. This means the light will turn off again.



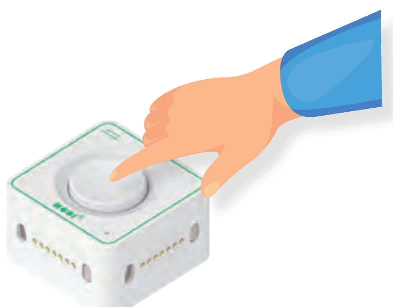
Do you remember which action Button can received?





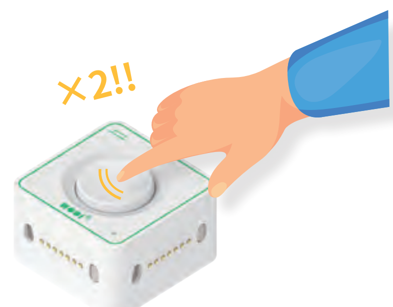
Button can receive 4 types of information.

1. CLICK



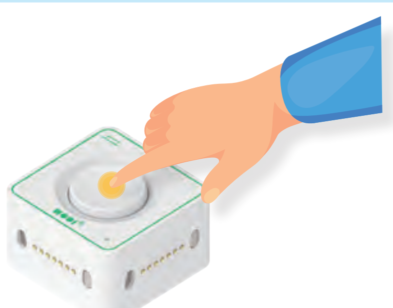
Detects if Button is clicked or not.

2. DOUBLE CLICK



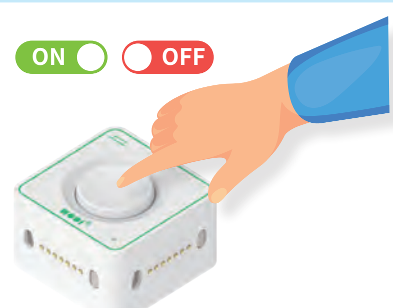
Detects if Button is double - clicked or not.

3. PRESS



Show the press status of Button.

4. TOGGLE



Changes on / off when Button is clicked.

Activity: Create Restroom Indicator

1 Materials Needed:

Network

Button

LED

2 Code story

When Button is Clicked

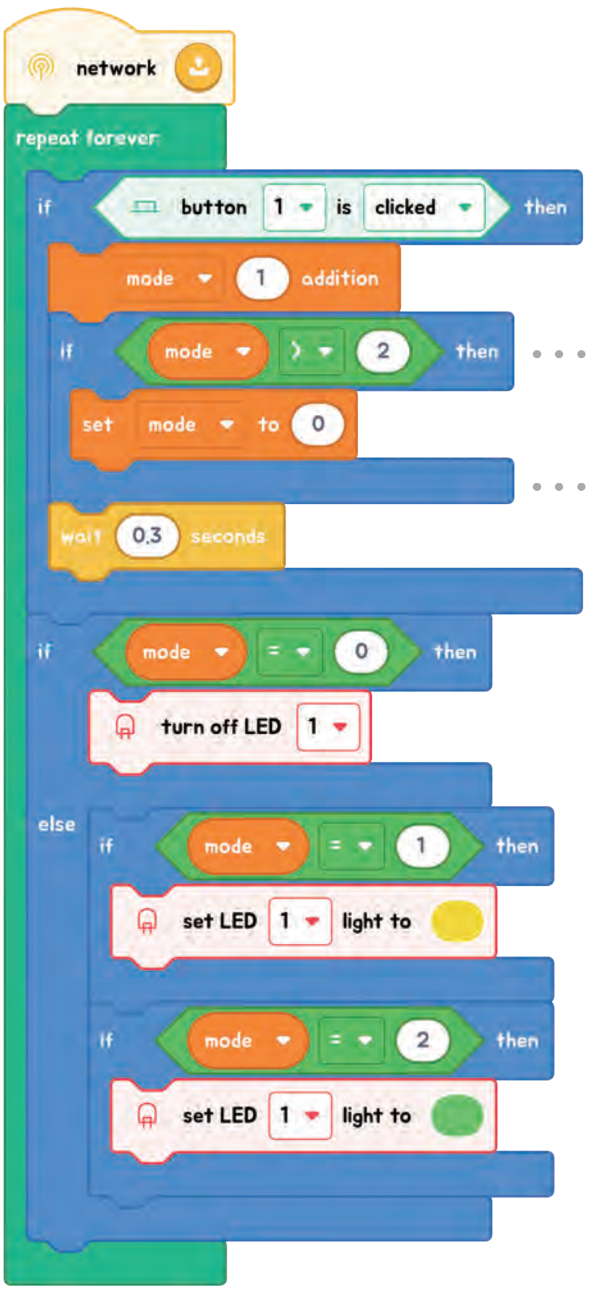
↓

add mode 1

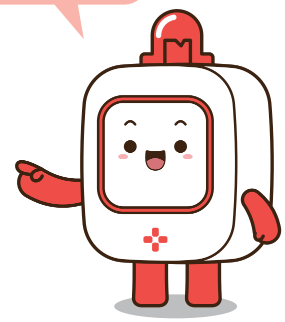
↓

if mode = 0	mode = 1	mode = 2
↓	↓	↓
turn off LED	LED as yellow	LED as green

Code Quest



When the mode is over than 2, set the mode to 0 because if the mode is 3 there isn't option to choose the light color.



Checkpoint

If you want to add different colors to the light, which block would you need to add?

The History of Lighting: From Fire to Electricity

Lighting has come a long way from the early days of human history. Let' s explore how people have used light through time and how it has changed to what we know today.

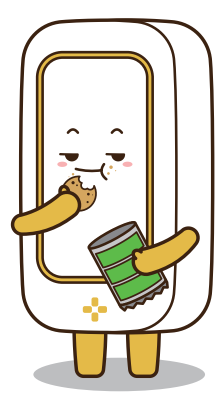
FIRE: THE FIRST LIGHT

A long time ago, before there were light bulbs or streetlights, people used fire as their primary light source. They would light torches or build campfires to see in the dark. This was important for cooking, staying warm, and staying safe at night.

Torches: These were sticks wrapped in cloth, dipped in oil, and then set on fire to create light.

Candles: Eventually, people learned to make candles from animal fat or wax. These burned for longer than torches and were safer to use indoors.

Why they were important: Oil lamps made it easier to keep a steady, bright light for a longer time without burning a big fire.



Oil Lamps

After fire and candles, people began using oil lamps. These lamps had a small container filled with oil and a wick that burned slowly. Oil lamps were common in homes and streets before electricity was invented.

The Invention of the Light Bulb

In 1879, Thomas Edison invented the electric light bulb, a significant moment in the history of lighting. Instead of fire or oil, the light bulb used electricity to create light. It was much safer, lasted longer, and could be easily turned on and off with a switch.

Impact of the light bulb: The invention of the light bulb changed everything! Streets, homes, and even schools became much brighter and safer. People could work or read even after the sun went down.

From the flicker of fire to the brightness of modern electric lights, how we light up our world has changed dramatically. It has made our lives safer, easier, and more fun!

Multiple Choice Questions

1 Which module can replace the Button we used in Mood Light?

Ⓐ Dial
Ⓑ Speaker
Ⓒ Display
Ⓓ Motor

CORRECT ANSWER Ⓐ

2 Why do programmers use variables in coding?

Ⓐ To change the colors of the screen
Ⓑ To store and reuse data in their programs
Ⓒ To make the program run faster
Ⓓ To play music in the background

CORRECT ANSWER Ⓑ

3 Which of the following is an example of an IoT device?

Ⓐ A regular bicycle
Ⓑ A paper notebook
Ⓒ A wooden chair
Ⓓ A smart fridge that can connect to the internet

CORRECT ANSWER Ⓓ

Reflection Questions

1 What are some other devices in your home that might use IoT?

2 How do you think IoT helps people in their daily lives?

3 What kind of light would you use for a birthday party, and why?

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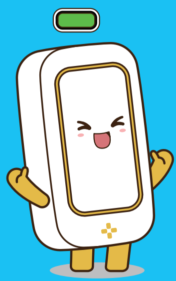

STEAM JOURNEY

EMBARK ON A CREATIVE EXPLORATION!

Welcome to "STEAM Journey with MODI," a groundbreaking textbook series designed to inspire the next generation of innovators, thinkers, and creators. Through the fascinating world of computing and robotics, intertwined with humanity, social sciences, health, and environmental studies, this series invites Grade 5 to 8 students to dive deep into the realms of Science, Technology, Engineering, Arts, and Mathematics.

WHAT'S INSIDE?

- Hands-On Learning: Engaging activities that blend theory with practical experiments, bringing concepts to life across STEAM disciplines.
- Broadened Perspectives: Insights into humanity and social sciences that highlight the ethical, cultural, and societal implications of technology.
- Real-World Applications: Discover how computing and robotics are transforming our world, from healthcare innovations to environmental solutions.
- Creative Problem-Solving: Foster critical thinking and creativity as students navigate through exciting challenges and projects, considering the broader human and ecological context.
- Inclusive STEAM Journey: Accessible content that encourages participation from all curious minds, regardless of their prior knowledge in technology, with a special emphasis on the interconnectedness of technological innovation and societal wellbeing.

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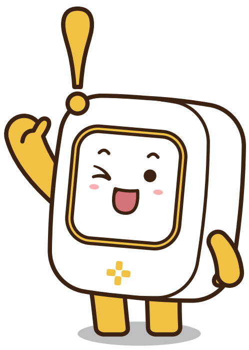
UNIT 1 CRACKING THE BINARY CODE

The Core of Digital Communication

Welcome to the intriguing world of binary code, the universal language of computers. Binary code uses just two symbols, 0 and 1, to represent all forms of data - from the text in an ebook to the colors in a photo. It's the simplicity of binary that makes it so powerful, allowing computers to perform complex operations with speed and precision.

Imagine telling secrets using only two words: "yes" and "no." In the world of computers, a similar language exists called binary code, made up entirely of two numbers: 1 and 0. Each '1'is like saying "yes" or "on," and each '0'is like saying "no" or "off." This simple system is how all digital devices talk and understand each other, from your computer and smartphone to even the microwave in your kitchen!



Understanding Binary

At its heart, binary is a way of representing information using only two options. Just as the alphabet forms the basis of written language, binary digits, or "bits," are the building blocks of digital communication. When bits are grouped together, they can represent anything from numbers and letters to colors and sounds, making up the complex data and instructions computers use to function.

HOW TO CALCULATE BINARY NUMBERS

Binary numbers work a bit like the counting numbers we use every day, but instead of going up to 10 before adding a new digit, binary only goes up to 2. For example, the binary number "10" isn't ten—it's two in our usual counting numbers because it represents "1 two and 0 ones."

HERE'S A QUICK WAY TO UNDERSTAND IT:

- The rightmost digit counts how many '1's there are.
- The next digit to the left counts how many '2's,
- The next after that counts '4's, then '8's, '16's, and so on, doubling each time.

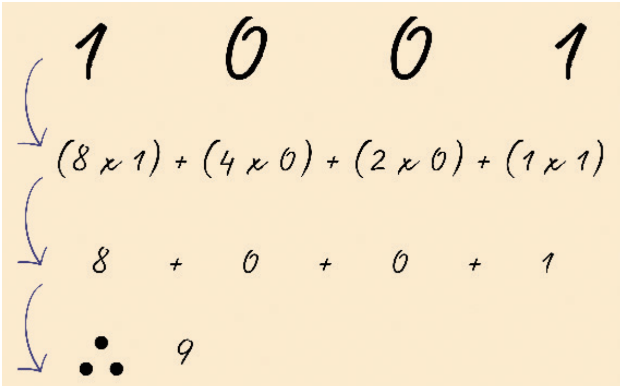
SO, THE BINARY NUMBER 1011 MEANS:

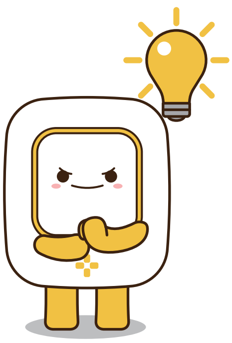
- 1 'eight', plus
- 0 'fours'(which means no fours),
- 1 'two', and
- 1 'one'.

Add them up (8 + 0 + 2 + 1), and you get 11 in our regular counting numbers!

LET'S TAKE A LOOK AT ANOTHER EXAMPLE! HERE, THE BINARY NUMBER 1001 MEANS:

- 1 'eight', plus
- 0 'fours'(which means no fours),
- 0 'twos', (which means no twos).
- 1 'one'.





Pop Quiz: Binary Number Calculation

QUIZ 1: BASIC BINARY CONVERSION

Convert the binary number 1010 to a decimal number.

A 10
B 8
C 5
D 12

ANSWER A

QUIZ 2: UNDERSTANDING BINARY PLACES

In the binary number 1101, what is the value of the digit in the second place from the right?

A 1
B 2
C 4
D 8

ANSWER C

QUIZ 3: BINARY TO DECIMAL

What decimal number does the binary number 0011 represent?

A 2
B 3
C 4
D 5

ANSWER B

QUIZ 4: COUNTING IN BINARY

Question: If you add 1 to the binary number 1111, what do you get?

A 10000
B 1110
C 10101
D 1000

ANSWER A

QUIZ 5: BINARY REPRESENTATION

Which binary number represents the decimal number 5?

A 101
B 110
C 111
D 100

ANSWER A

QUIZ 6: DECODING BINARY

What decimal number is represented by the binary number 1001?

A 8
B 9
C 10
D 11

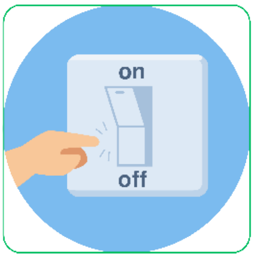
ANSWER B

BINARY IN EVERYDAY LIFE

To grasp binary, let's relate it to something familiar: a row of light switches. Each switch, like a bit, has two possible states. If we consider a row of three switches, each can be either up (1) or down (0). This setup can represent different values, much like how bits in a computer represent data.
For instance, all switches up (111) could represent the number 7, while all down (000) could represent 0. Below are other examples of the secret language of all our digital gadgets in daily life.

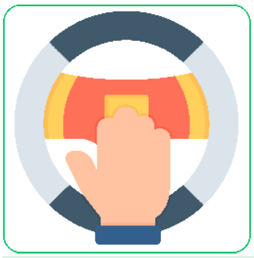
LIGHT SWITCHES

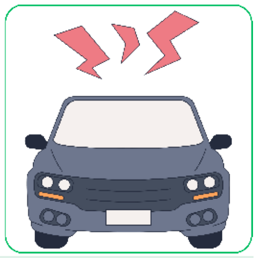
A light switch is a perfect real-life example of binary. When you flip a switch up, the light turns on (1), and when you flip it down, the light turns off (0). Just like in binary code, where 1 represents 'on'and 0 represents 'off,'the light switch only has these two states.



CAR HORNS

Think about using the horn in a car. You press it (1), and it makes a sound. You release it (0), and it stops. This on/off system is binary in action. The horn doesn't care about 'how'you press it, only whether it's being pressed or not, much like a binary digit being set to 1 or 0.



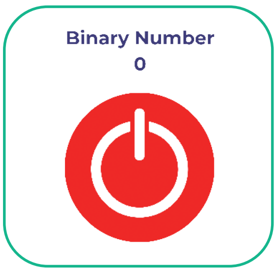


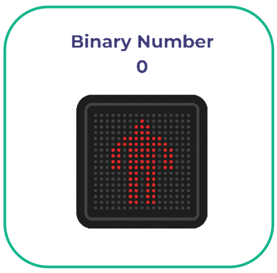
POWER BUTTONS

The power button on many devices, from your computer to your television, works in a binary manner. Press it once to turn the device on (1), and press it again to turn it off (0). Some devices even show this with symbols: a line for 'on'(1) and a circle for 'off'(0).

CROSSWALK SIGNALS

Crosswalk signals at intersections are another example. They're either in a 'walk'state (1), allowing pedestrians to cross, or a 'don't walk'state (0), indicating to wait. This binary system helps manage pedestrian traffic safely and efficiently.





Reflection Section

MULTIPLE CHOICE QUIZZES

1 **What is the basic language of computers, using only two digits?**

- (A) Decimal
- (B) Hexadecimal
- (C) Binary
- (D) Morse code

CORRECT ANSWER (C)

2 **Binary code consists of which two digits?**

- (A) 1 and 2
- (B) 0 and 1
- (C) A and B
- (D) 0 and 9

CORRECT ANSWER (B)

Writing Questions Requiring Critical Thinking

Binary Code in Daily Life

Think about gadgets you use every day, like a calculator, computer, or game console. Write a few sentences about how you think binary code helps these gadgets work. For example, how does binary code help a calculator add numbers or a game console play your favorite game?

Explaining Binary to a Friend

Imagine your friend has never heard of binary code and is curious about how computers use just "0" and "1" to do so many things. Write a short explanation to help your friend understand. You can compare it to something simple, like an on-off switch, to make it easier to grasp.